Initial Conditions and the Big Push*

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Abstract

This paper examines how initial conditions shape the long-run impact of large temporary shocks on economic geography. Using WWII supply contracts in the United States, we show that temporary spending had persistent effects on population and manufacturing employment only in counties with low initial manufacturing development. Motivated by these findings, we develop a multisector economic geography model that features external economies of scale in manufacturing with non-constant elasticity. We calibrate the model and show that we match the empirical estimates with an S-shaped scale function for which scale economies large enough to cause coordination failures are limited to locations with low initial manufacturing employment. We use the calibrated model to re-evaluate how temporary WWII spending shaped the evolution of U.S. postwar economic geography. Despite the presence of big push dynamics and multiple steady states, the aggregate effects of WWII spending on long-term outcomes are limited due to the restricted domain over which coordination failures operate.

JEL Classification: N40, N92, O14, O40, R11.

Keywords: WWII spending, economic geography, path dependence, external economies, big push.

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1 Introduction

A central question in economic development is whether temporary "big push" interventions can trigger lasting industrialization, or whether spatial patterns simply revert to fundamentals. Models of multiple steady states imply that transitory shocks can permanently reallocate economic activity by overcoming local coordination failures. Yet the empirical literature remains divided between studies finding persistent lock-in and those finding rapid mean reversion.[1] A key limitation of this debate is that it treats path dependence as an all-or-nothing property of the economy, largely overlooking the core theoretical prediction that the long-run response to a temporary shock depends systematically on a location's initial condition.

This paper examines this question using one of the largest temporary fiscal shocks in U.S. history: federal spending during World War II (WWII). During the war, the U.S. government undertook an unprecedented public spending program, with federal net outlays rising from roughly 10% of gross domestic product (GDP) to over 40% in just a few years, then subsequently retreating to less than 15% (Figure 1). Much of this increased spending consisted of supply contracts awarded to domestic firms for the production of materials and equipment for the war effort. To evaluate the effects of this spending on local outcomes, we link a comprehensive dataset of WWII war supply contracts with data on the long-run growth of U.S. counties from 1930 to 2000.

Using a potential outcomes framework alongside controls for initial conditions, we compare the long-run trajectories of counties that received large war supply contracts relative to their prewar manufacturing employment to those that received small or no contracts. Consistent with the prior literature, we find that treated counties experienced higher overall long-run population and manufacturing employment growth. Our primary empirical contribution, however, is to show that these positive treatment effects

[1] A few examples from a vast literature include Bleakley and Lin (2012); Davis and Weinstein (2002); Juhász (2018); Lin and Rauch (2022).

are driven almost entirely by counties with initially small manufacturing employment. While these less-industrialized counties experienced economically large and statistically significant long-run increases in economic activity, heavily industrialized counties saw their substantial wartime gains almost completely reverse in subsequent decades.

We interpret our empirical results as providing support for "big push" or "poverty trap" theories of regional development (Rosenstein-Rodan, 1943; Murphy, Shleifer, and Vishny, 1989; Krugman, 1991; Azariadis and Stachurski, 2005). In these frameworks, coordination failures can trap a region in an equilibrium characterized by low economic activity relative to its potential. A sufficiently large temporary shock can transition the region to a higher, self-sustaining equilibrium. However, as the term "poverty trap" implies, these theories posit some scale of economic activity beyond which coordination failures vanish and temporary shocks no longer yield permanent effects. While these theories have motivated a large empirical literature on path dependence, the prediction of heterogeneity in the long-run responses based on initial conditions has been largely neglected. It is precisely this prediction of the limited relevance of coordination failures beyond a certain industrial scale that we find support for in this paper.

Motivated by these findings, we develop a quantitative multi-region economic geography model that accommodates coordination failures and path dependence, but allows the domain of these failures to vary with the scale of economic activity. We extend the recent framework of Allen and Donaldson (2020) by introducing multiple sectors to capture long-run structural change, and, crucially, by incorporating local external economies of scale in manufacturing with non-constant elasticity. We calibrate this scale function so that the heterogeneous responses to temporary government demand shocks from the model match our empirical estimates. We find that the model fits the data well with an S-shaped scale function in manufacturing employment: it is initially flat up to 250 workers, increases with an elasticity of about 0.2 up to 1,500 workers, and becomes flat again thereafter. The calibrated model generates multiple interior steady

states and path dependence. In contrast, the standard constant-elasticity model ubiquitous in the literature fails to match our empirical moments because it lacks a mechanism to restrict the domain over which coordination failures apply.

We use the calibrated model to re-evaluate how wartime expenditures shaped the long-run evolution of U.S. economic geography. We simulate the model with and without the WWII spending shocks and compare spatial outcomes in the year 2000. We find that despite substantial reallocation of population and manufacturing employment across smaller counties, the WWII shocks had minimal long-term effects on aggregate U.S. manufacturing employment, welfare, or broader regional reallocation. This occurs because the restricted domain of coordination failures limits the scope for spatial reallocation to raise aggregate productivity. Because the S-shaped scale function flattens above 1,500 workers, counties that undergo a big push quickly exhaust their capacity to generate further external economies of scale. Consequently, the permanent long-run gains in treated small counties are offset by the contraction of other small, untreated counties. In contrast, the large initial scale of industrial activity in heavily industrialized counties places them beyond the domain of coordination failures, so that their temporary wartime gains are not sustained and thus do not generate much long-run reallocation or aggregate growth.

Our paper contributes to three strands of literature. First, we add to the extensive empirical literature on historical persistence and path dependence in economic geography (Rauch, 1993; Bleakley and Lin, 2012; Davis and Weinstein, 2002; Juhász, 2018; Lin and Rauch, 2022). The theoretical prediction that persistence is heterogeneous with respect to initial conditions has been largely ignored empirically, and our main contribution to this literature is to document and analyze this state-dependence. Second, we contribute to the recent literature on quantitative models of multiple steady states and "big push" dynamics (Allen and Donaldson, 2020; Garg, 2025; Buera, Hopenhayn, Shin, and Trachter, 2021) by developing a tractable multi-region model that embeds big push

dynamics and explicitly parameterizes the spatial domain over which they operate. Finally, we make a contribution to the empirical literature on the effects of WWII spending (Fishback and Cullen, 2013; Fishback and Jaworski, 2016; Jaworski, 2017; Rhode, Snyder Jr, and Strumpf, 2018; Li and Koustas, 2019; Garin and Rothbaum, 2025; Jaworski and Yang, 2025), which generally documents modest positive long-run impacts on local economies. Relative to prior research that examines wartime investments in long-lived structures (Bateman, Ros, and Taylor, 2009; Garin and Rothbaum, 2025; Jaworski and Yang, 2025), we focus on the transient component of the shock and its interaction with pre-existing industrial structures. Our findings show that the long-run aggregate impact of this temporary spending was rather limited, despite substantial local effects in some areas.

The rest of the paper proceeds as follows. Section 2 briefly describes the data and the historical context. Section 3 presents the empirical results. Section 4 describes the model and the structural estimation. Section 5 conducts counterfactual analysis, and Section 6 concludes.

2 Estimation and Identification

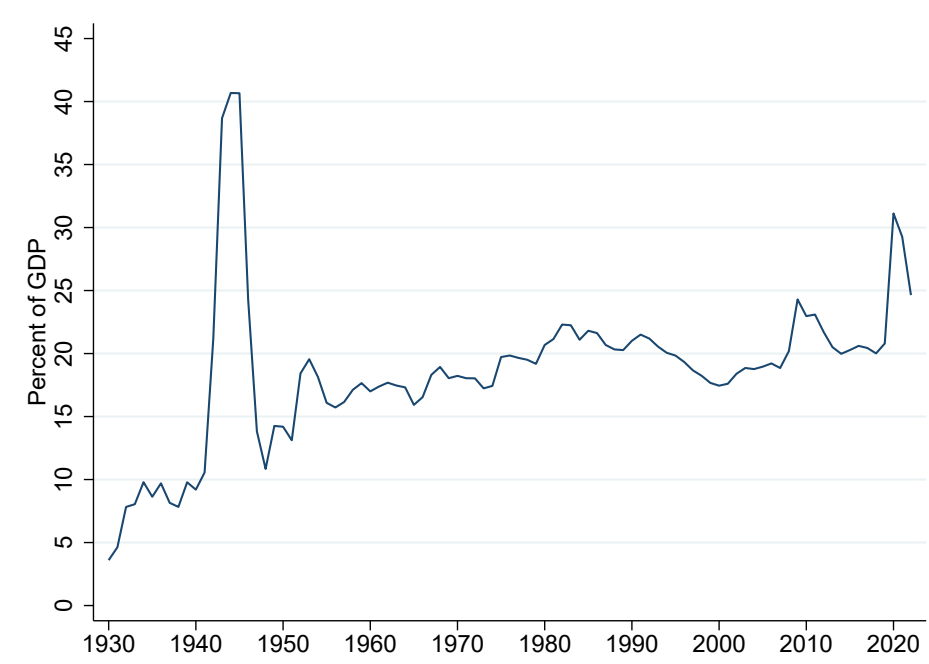
This section briefly describes the historical context of federal military supply contracts during WWII and our data sources, then discusses our approach to estimation and identification.

2.1 Historical Context and Data

During WWII, the federal government assumed an unprecedented level of control over the U.S. economy. At their peak, federal government expenditures accounted for over 40 percent of GDP, the highest proportion ever recorded (Figure 1). The federal government allocated $196 billion for military supply contracts (primarily for manufactured

goods) and invested an additional $31 billion in new production facilities between 1940 and 1945.[2] This remarkable endeavor to turn the U.S. into the "Arsenal of Democracy" stands as the largest economic intervention by the federal government in U.S. history.

Figure 1. Federal Net Outlays as Percent of GDP



Note: This figure shows the federal net outlays as percent of GDP over time. Source: Federal Reserve Bank of St. Louis, U.S. Office of Management and Budget; see the link here.

The allocation of war supply contracts during WWII was primarily determined by economic considerations rather than political factors. Rhode et al. (2018) find that the strongest predictors of the assignment of supply contracts at the state level were existing general industrial capacity prior to the war in 1939 and specialized industrial capacity for aircraft production. Their findings suggest that the government prioritized regions with established industrial infrastructure and specialized capabilities, reflecting a strategic approach to efficiently meet the demands of wartime production (Fishback and Cullen, 2013). The ability of the United States to rapidly procure and produce goods and services contributed to the Allied victory in WWII (Harrison, 1998).

[2]For reference, U.S. GDP was about $102 billion in 1940.

The data for WWII supply contracts are sourced from volumes originally issued by the Civilian Production Administration (formerly the War Production Board), which recorded all defense contracts greater than $50,000 issued between June 1940 and September 1945. These military supply contracts were awarded for the production of combat supplies (e.g., planes, tanks), intermediate goods (e.g., propellers, gun fittings), and incidental materials (e.g., clothing, medical supplies). War spending was divided into two broad categories: supply contracts and facilities spending. Facilities spending, which accounted for approximately 13% of total wartime expenditures, financed long-lived structures such as industrial plants, shipyards, air fields, and cantonments. We explicitly focus on supply contracts and exclude facilities spending for two key reasons. First, supply contracts represent the vast majority of total war spending. Second, and crucially for our empirical design, procurement contracts capture a sharp, temporary demand shock, whereas physical structures and capital facilities depreciate slowly over long horizons.[3]

We construct our measure of local government spending by aggregating the monetary value of these military contracts with private firms at the county level. The dataset comprises over 191,000 contracts with a total value of $183 billion.[4] The distribution of contract values was highly skewed; while more than 90% of contracts were for less than $1 million, a small number of high-value awards, particularly in aircraft production, accounted for a large share of total spending. For instance, major contracts included a $669 million award to Boeing for B-17 bombers in Seattle, WA, and engine contracts with Pratt & Whitney ($403 million in East Hartford, CT) and Wright Aeronautical ($429 million in Lockland, OH). Furthermore, these procurement agreements were intentionally short-term, a design intended to mitigate the risk of renegotiation under wartime inflation. The median contract duration was seven months, with 90% of all contracts

[3]For studies focusing on the long-run effects of WWII government-financed facilities and capital investments, see Fishback and Cullen (2013), Garin and Rothbaum (2025), and Jaworski and Yang (2025).

[4]This figure represents 93% of the $196 billion in contracts awarded, with the remainder falling below the $50,000 reporting threshold.

having a duration of 14 months or less. This short-term nature makes the contract data particularly suitable for measuring a temporary fiscal shock.

Data on county characteristics are the same as those in Kline and Moretti (2014) and come from a variety of sources including the Census of Population and Housing and Agricultural Census. We use the crosswalk provided by Eckert, Gvirtz, Liang, and Peters (2020) to obtain a sample of 3,141 harmonized counties between 1900 and 2000. Online Appendix B documents these steps.

2.2 Empirical Model

Our goal is to estimate the impact of this massive but temporary increase in government spending on local economic outcomes in the long run, and, in particular, to understand how this impact varied with the initial size of the local manufacturing sector. Our motivation for exploring this potential heterogeneity comes from big push or poverty trap theories of regional specialization and development (Rosenstein-Rodan, 1943; Murphy et al., 1989; Krugman, 1991; Azariadis and Stachurski, 2005), in which coordination failures can result in a region being stuck in an equilibrium that features a low level of economic activity relative to its potential. Large temporary shocks to underdeveloped regions can induce a higher level of activity, which then becomes self-sustaining and permanent. This implication has been borne out in previous research on WWII spending and in many (though not all) other contexts.[5]

We focus on a second important implication: temporary shocks can only have permanent effects on regions trapped in a low-activity equilibrium. For regions that are already highly developed, even a very large temporary shock will not affect their long-term trajectory. This theoretical prediction of heterogeneity in the long-run response based on initial conditions has seen little or no empirical investigation, and is the focus

[5]For the U.S. experience in WWII, see, for example, Fishback and Jaworski (2016); Li and Koustas (2019). Examples of studies in other contexts include Bleakley and Lin (2012); Kline and Moretti (2014); Juhász (2018).

of our analysis in this paper. Following a long tradition in the literature, we consider manufacturing as the sector generating increasing returns during this time period, and hence the size of the manufacturing sector as the key initial condition for understanding heterogeneous responses to shocks.[6]

We draw on several additional implications of the theory to inform our empirical analysis. Since an underdeveloped manufacturing sector is a stable equilibrium, small shocks will have no long-lasting effect even in underdeveloped regions. On the other hand, once a shock is large enough to move the region onto a trajectory leading to the higher equilibrium, further increases in the size of the shock will not alter the long-run outcome. These considerations suggest dichotomizing the war spending variable based on a threshold, dividing regions (counties in our analysis) into "treated" (large shocks) and "untreated" (small or no shocks) categories for the analysis. Similar reasoning suggests dichotomizing the initial conditions, dividing counties based on high and low initial manufacturing employment (our measure of sector size). We classify observations into four quadrants using these criteria and then compare long-run outcomes in the treated quadrants to their untreated counterparts.

To be clear, the conditional expectation of the treatment effect should vary continuously with both the initial conditions and the treatment size.[7] Our discretization strategy is dictated by the need to maximize the relevant information that can be obtained from limited data. Rather than estimate the entire two-dimensional surface of treatment effects, which is infeasible to do with any precision, we average the treatment effects over the most relevant domains by choosing appropriate thresholds that define the

[6] See Lashkaripour and Lugovskyy (2023) and Bartelme, Costinot, Donaldson, and Rodriguez-Clare (2025) for some recent evidence on scale economies in manufacturing.

[7] Continuity is expected because the treatment size and initial manufacturing employment are not sufficient to identify which counties will be deflected from their counterfactual untreated long-run equilibrium trajectory by the treatment. For example, some counties with low initial manufacturing employment will already be on a trajectory towards a high equilibrium and will thus have a treatment effect of zero for even large treatment sizes, even though most counties with the exact same initial manufacturing employment will have positive treatment effects for most treatment sizes. Integrating over unobserved heterogeneity leads to continuity of the conditional expectation despite the discrete nature of the transition from a low to high equilibrium.

quadrants. The choice of thresholds is guided by both substantive and statistical considerations; we discuss our choices in more detail and conduct robustness checks in the following section, but we stress that there is no right or wrong way to choose these cutoffs, only more or less informative ones. We do not interpret the treatment effects as structural parameters of an underlying model. We will use the estimated treatment effects as moments to be targeted when estimating our structural model in Section 4.

We follow Kline and Moretti (2014) and adopt the potential outcomes framework as our empirical model, with the average treatment effect on the treated (ATT) as our primary parameter of interest.[8] If county i actually received treatment, its treatment effect is the difference between its actual outcome and the (unobserved) counterfactual outcome had it not received treatment. The ATT is simply the average of these individual treatment effects taken over all counties that actually received treatment. Formally, let y_i^0 be the outcome of interest (e.g., the long-run change in manufacturing employment) in county i *if it had not received treatment*. We assume that

$$y_i^0 = \alpha_i^0 + \theta_0 X_i + \epsilon_i^0, \tag{1}$$

where X_i is a set of observable control variables and ϵ_i^0 and ϵ_i^1 are unobserved terms. If county i *were to receive treatment*, the outcome y_i^1 would be given by

$$y_i^1 = \alpha_i^1 + \theta_1 X_i + \epsilon_i^1. \tag{2}$$

Identification is achieved by assuming that both residuals are uncorrelated with the

[8] See Imbens and Wooldridge (2009) for a review of the literature on estimation and inference in this model. The ATT can be consistently estimated under weaker assumptions than the average treatment effect (ATE). It also has an appealing interpretation in historical counterfactual analysis, especially when general equilibrium effects are important, since it avoids considering very different policies than those actually implemented.

treatment conditional on the controls,

$$E[\epsilon_i^0|X_i, D_i] = E[\epsilon_i^1|X_i, D_i] = 0, \ \forall i. \tag{3}$$

The treatment effect for county i is given by $\alpha_i^1 - \alpha_i^0 + (\theta_1 - \theta_0)X_i$, and the ATT for each subgroup is obtained by averaging across treated counties in each subgroup. We allow for the possibility that the treatment effect is correlated with the observables, i.e., $\theta_0 \neq \theta_1$; if $\theta_0 = \theta_1$, the model collapses to a standard difference-in-differences regression. Equations (1) and (2) are estimated separately on the untreated and treated populations, respectively, and the estimates are then combined to compute the ATT.[9]

The model (1)–(3) can be regarded as a local approximation to a structural model, such as the one we develop in Section 4. The controls X_i should include observables from the initial state that summarize all aspects of the local initial conditions useful in predicting the future evolution of local productivity and amenities. They should also account for WWII spending in nearby counties, which will affect outcomes in a treated county through spatial spillovers, such as those generated by trade and migration frictions. Given our focus on the temporary component of WWII spending, these controls should also include wartime government investment in long-lived capital assets such as new factories. Conditional on the appropriate set of controls, the residuals reflect the portion of future local productivity or amenity growth unforecastable based on the initial state. The identification assumption is that the local WWII supply spending is uncorrelated with this unpredictable component of local productivity or amenity growth.

This description raises the question of where our identifying variation comes from, given that treatment is non-randomly assigned and that we control for many of the observables in the initial state. We consider our identifying variation as coming from three primary sources. One is the composition of local industry: two places with the same

[9] This is known as the Oaxaca-Blinder estimator (Oaxaca, 1973; Blinder, 1973). See also Kline (2011) for some robustness properties of the estimator and Kline and Moretti (2014) for an application in a similar context to ours.

level of manufacturing employment may have very different wartime demand depending on how useful the products they specialize in are for the war effort (see Rhode et al. (2018) for some evidence). Since the composition of wartime demand differs markedly from the composition of peacetime demand, this variation is plausibly exogenous with respect to the future evolution of local productivity and amenities. The second source of variation comes from political, social or informational connections between wartime decisionmakers and particular geographic locations or firms, which again is plausibly exogenous with respect to long-run changes in fundamentals. The third source is the winner-take-all nature of most contracts, which implies that there might be very little difference between a county that received a large contract and one that did not. This last source could be particularly influential for counties with an initially small manufacturing base, for which the allocation of a single large contract could have a material effect on output and employment.

Admittedly, it is difficult to be completely confident that there is no confounding selection into treatment whenever the treatment is not explicitly randomly assigned. We address this concern, as best we are able, with two data exercises and a theoretical point. First, we use a placebo test to show that treatment status does not predict outcomes in the pre-treatment period. Second, we use data on industry composition to show that our results are robust to controlling for this source of variation. Finally, our central estimand is the *difference* in the ATT between locations with initially low and initially high manufacturing employment. That difference is identified under the weaker assumption that selection bias affects both counties equally—a considerably weaker requirement than unbiased selection into treatment overall.

3 Empirical Results

Treatment and Initial Conditions We split the set of counties into those with an initially high manufacturing base (1930 log manufacturing employment above the 50th percentile nationally) and those with an initially low manufacturing base. We assign treatment status to each county based on the WWII war supply contract value per 1930 manufacturing employee. Counties with a high level of war supply contracts were more densely populated and had higher manufacturing employment in the prewar period (see Table A.1 for summary statistics and Figure A.1 for the distribution of initial manufacturing employment). This is not surprising because war supply contracts were allocated based primarily on existing production capacity (Fishback and Cullen, 2013).

We choose the definition of treatment so as to balance the dual objectives of ensuring an adequate number of control units for both groups and having the treatment group assignment lead to similar distributions of contract value within the treatment and control groups, across initial conditions. Our baseline is to assign counties to the treatment group if their log contract value per 1930 manufacturing employee exceeds 8, a round number very close to the mean. We conduct extensive robustness checks on both cutoffs (initial conditions and treatment) and discuss them further below.

Pretrend Test To adjust for pre-existing differences across counties before the WWII, we follow previous literature and control for a rich set of county-level variables related to prewar economic, social, demographic, and geographical characteristics in 1930 (Kline and Moretti, 2014; Li and Koustas, 2019). The covariates include log population, log employment, log manufacturing employment, agricultural employment share, manufacturing employment share, log manufacturing wages, dummies for wages in manufacturing or trade being missing, farm values, urban population share, the share of white population, the share of the population age 5 or above that are illiterate, the share of whites who are foreign-born, the share of households with a radio, maximum elevation,

elevation range, and area.[10]

Since contract assignment is strongly correlated with initial manufacturing conditions (Rhode et al., 2018), we control for relevant variables in the manufacturing sector including log total manufacturing employment, manufacturing employment share, and log manufacturing wages. We control for an indicator variable for whether a region is classified as having initially high manufacturing employment, as well as an interaction of this indicator with the log prewar population, log prewar manufacturing employment, and the prewar manufacturing share. We also control for trends in population and manufacturing employment between 1900 and 1920. To account for potential spillover effects from neighboring counties, we control for the war supply contract value per capita of neighboring counties. We also control for WWII facilities spending to purge any lasting direct effects of long-lived structures on productivity. To better balance the treated and control groups, we drop counties whose pre-WWII characteristics yield extreme predicted probability of treatment higher than 99% or lower than 1%.

We first assess whether the set of control variables performs well to adjust for observable differences across counties. To do so, we conduct a placebo test by estimating the "effects" of defense spending on 1900-1930 changes in population and manufacturing employment, for all counties together as well as separately for initially low and high manufacturing counties. This test attempts to detect if, conditional on the control variables, the outcome variables exhibit differential trends before WWII across treated and control counties. Because that time period predates the war, a finding of a significant relationship would signal a potential failure of our identification assumption. As shown in Table 1, no statistically or economically significant effects are discernible across all outcomes and subgroups.

[10]To capture potential temporal correlation of shocks, we control for socioeconomic characteristics in 1930 and 1900 (except log population and log manufacturing employment in 1900 to prevent multicollinarity, as we also control for trends in these variables).

TABLE 1. Pretrend Tests of Outcomes in 1900-1930

	(1) Pop 1900-1930	(2) ManufEmp 1900-1930	(3) Pop 1900-1930	(4) ManufEmp 1900-1930	(5) Pop 1900-1930	(6) ManufEmp 1900-1930
Treat	−0.001	0.004	−0.004	−0.008	0.000	0.008
	(0.004)	(0.007)	(0.004)	(0.015)	(0.005)	(0.008)
Location	All	All	LowM	LowM	HighM	HighM
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows pretrend tests for the "treatment" effect on the growth rate between 1900 and 1930 (defined as the log difference between 1930 and 1900 levels divided by three) of the outcome variables. Columns (1)–(2) show pretrend tests for the whole sample, columns (3)–(6) break them down by the initial levels of manufacturing employment: whether a county has manufacturing employment in 1930 below or above the 50th percentile. Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms. Block-bootstrapped standard errors at the state level are in parentheses.

Main Results Panel (a) of Table 2 shows the results of estimating equations (1) and (2) and computing the ATT across all observations, regardless of initial conditions. Columns (1)–(2) report the long-run effects on population and manufacturing employment growth. Between 1930 and 2000—five and a half decades after the war ended—treated counties grew 1.0% and 1.6% faster per decade, respectively. These modest positive long-run average effects are consistent with the previous literature on the geographic impact of WWII spending (Fishback and Cullen, 2013; Fishback and Jaworski, 2016; Li and Koustas, 2019). For the immediate postwar period, up until 1960, we see much larger differential growth rates of 2.3% and 6.0% for population and manufacturing employment in treated counties (columns (3)–(4)). However, these sizable impacts die out or are partially reversed over the next several decades (columns (5)–(6)). This dynamic pattern of growth and (partial) reversal reflects the temporary nature of the WWII demand shock and the gradual adjustment of treated counties to the loss of wartime demand for their products.

Panel (b) of Table 2 reports the ATT separately for counties with high (*HighM*) and low (*LowM*) initial manufacturing bases. For regions with initially low manufacturing

TABLE 2. Long-Term Impact on Growth Rate of Outcomes

Panel (a): Baseline

	(1) Pop 1930-2000	(2) ManufEmp 1930-2000	(3) Pop 1930-1960	(4) ManufEmp 1930-1960	(5) Pop 1960-2000	(6) ManufEmp 1960-2000
Treat	0.010	0.016	0.023	0.060	0.002	−0.017
	(0.006)	(0.007)	(0.008)	(0.012)	(0.006)	(0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Panel (b): Initial Condition

	(1) Pop 1930-2000	(2) ManufEmp 1930-2000	(3) Pop 1930-1960	(4) ManufEmp 1930-1960	(5) Pop 1960-2000	(6) ManufEmp 1960-2000
Treat*LowM	0.029	0.038	0.029	0.087	0.028	−0.003
	(0.008)	(0.010)	(0.009)	(0.019)	(0.009)	(0.010)
Treat*HighM	0.005	0.010	0.021	0.053	−0.004	−0.021
	(0.007)	(0.008)	(0.009)	(0.013)	(0.007)	(0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows the impact of wartime spending on the growth rate of outcomes. Columns (1)–(2) show the treatment effect on the growth rate of outcome variables between 1930 and 2000 (defined as the log difference between 2000 and 1930 levels divided by seven). Columns (3)–(6) show the treatment effect on the growth rate of outcome variables by period: 1930-1960 and 1960-2000. Panel (a) shows the treatment effects for the whole sample, and panel (b) breaks them down by the initial levels of manufacturing employment: whether a county has manufacturing employment in 1930 below or above the 50th percentile. Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms. Block-bootstrapped standard errors at the state level are in parentheses.

employment, wartime defense spending increased the long-term growth rate of population and manufacturing employment levels by 2.9% and 3.8%, respectively, between 1930 and 2000. For regions with initially high manufacturing employment, these effects are much smaller both economically and statistically: the long-run effects are 0.5% and 1.0%, respectively, with both statistically indistinguishable from zero. In economic magnitude, the differential impact on manufacturing employment between the two groups of regions of 2.8% per decade over seven decades is about 20 log points, so a treated low-manufacturing county had roughly 22% more manufacturing employment in 2000 than its high-manufacturing counterpart.

Columns (3)-(6) report the dynamic patterns, which show that both treated groups

experienced similar large positive effects on both population and manufacturing between 1930 and 1960. However, these positive impacts almost completely reverse over the next 40 years for the initially high manufacturing group, leaving the long-term impact for this group close to zero. In contrast, the initially small treated locations experienced little or no reversal from 1960-2000, leaving their population and manufacturing employment materially higher after 60 years.

Our results show that the strong positive long-run effects of WWII spending on population and manufacturing employment are found only for regions with initially small manufacturing sectors, as predicted by big push/poverty trap models. The dynamic pattern of the effects also supports the big push theory. The short-term effects are similar for both groups, since we would expect a large temporary expansion in employment due to the demand shock regardless of any endogenous productivity effect. However, as the demand shock recedes only the locations that have been pushed to a higher productivity state retain sizable employment gains, while they largely fade away for locations that were not.

Treatment Intensity One may wonder whether the long-run effects simply reflect the size of the wartime treatment rather than a discrete transition between equilibria. Table 3 breaks the treatment size into two categories, with the log contract value per 1930 manufacturing employment above or below median, and re-estimates baseline regression equations (1) and (2). The results are as expected: larger treatments lead to larger short-term effects, reflecting the larger direct effect, and also larger long-term effects, reflecting the higher probability of being pushed onto a higher productivity trajectory. The long-run effects continue to be much larger for the group with the initially smaller manufacturing base.

Varying the Cutoffs There may be concerns that the results are artifacts of the particular cutoffs we chose for the initial-condition split and the treatment threshold. Figure

TABLE 3. Long-Term Impact on Growth Rate of Outcomes by Treatment Size

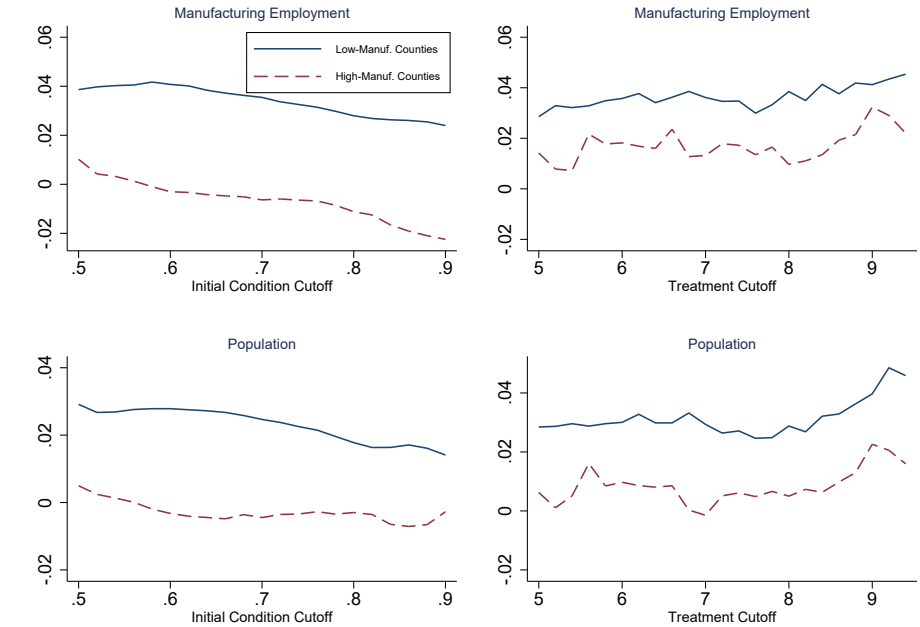
	(1) Pop 1930-2000	(2) ManufEmp 1930-2000	(3) Pop 1930-1960	(4) ManufEmp 1930-1960	(5) Pop 1960-2000	(6) ManufEmp 1960-2000
Treat Big*LowM	0.050	0.052	0.042	0.117	0.056	−0.011
	(0.017)	(0.016)	(0.019)	(0.022)	(0.015)	(0.017)
Treat Small*LowM	0.014	0.029	0.021	0.067	0.009	0.001
	(0.008)	(0.011)	(0.011)	(0.022)	(0.010)	(0.013)
Treat Big*HighM	0.014	0.022	0.038	0.091	−0.001	−0.025
	(0.009)	(0.013)	(0.013)	(0.022)	(0.008)	(0.011)
Treat Small*HighM	−0.000	0.001	0.009	0.027	−0.007	−0.017
	(0.007)	(0.008)	(0.008)	(0.011)	(0.007)	(0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows the impact of wartime spending on the growth rate of outcomes by treatment size. Columns (1)–(2) show the treatment effect on the growth rate of outcome variables between 1930 and 2000 (defined as the log difference between 2000 and 1930 levels divided by seven). Columns (3)–(6) show the treatment effect on the growth rate of outcome variables by period: 1930-1960 and 1960-2000. Low and high initial manufacturing base is defined by being above or below the 50th percentile of manufacturing employment in 1930. Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms. The boundary between "small" and "large" treatment is defined by a log per 1930 manufacturing employee supply contract value of 9.36, which is the median value in the treated group. Block-bootstrapped standard errors at the state level are in parentheses.

2 shows how varying the cutoffs for the initial high-low split as well as the treatment size affects the results. Classifying more locations as "small" and fewer as "large" results in a monotonic decline in both coefficients, with the effects for large locations turning negative as the category comes to include only the very largest locations. The spread between the estimates is fairly robust. These results suggest that it is indeed the observations in the smaller percentiles of the initial conditions that are driving the positive long-run effect. The results are also not too sensitive to the definition of treatment, as is also shown in Figure 2.

Industrial Composition As mentioned above, our identifying variation comes from three primary sources: the composition of local industry, locally specific political and informational connections, and true randomness from the lumpy and winner-take-all nature of the contracts. The only observable element is the composition of local indus-

Figure 2. Coefficients with Varying Cutoffs of Treatment and Initial Conditions



Notes: This figure shows the coefficients as the cutoffs of treatment and initial condition definitions vary. The left panel corresponds to estimates as the cutoff of classifying initially high- and low-manufacturing locations ranges from 50th to 90th percentiles. The right panel corresponds to estimates as the treatment cutoff varies from 4 to 10 in log contract value per 1930 manufacturing employee.

try, and we now explore how using (or not using) this information affects our estimates. We divide manufacturing sectors into those highly related to the war effort (e.g., auto, ship and aircraft manufacturing) and all others. Table A.2 shows that the share of prewar manufacturing employment in these sectors is highly correlated with both treatment status and contract value. This composition may also be related to the future evolution of productivity and amenities, despite our other controls, which would violate our exclusion restriction.

The concern is sharpest for our heterogeneity result: manufacturing sub-industries may differ in their scale elasticities, and larger wartime centers specialized in aircraft, shipbuilding, and munitions, which contracted sharply after 1945. Smaller counties, by contrast, may have been better positioned in general-purpose or dual-use goods, which were less susceptible to these boom-and-bust cycles. If contract intensity is cor-

related with preexisting specialization, then stratifying by total manufacturing employment could confound composition with agglomeration.

We run three robustness tests to resolve these issues. First, Table A.3 re-runs our basic specification while controlling for industry composition. Reassuringly, the results are very similar to those reported in Table 2. Second, we replace actual contract allocation with initial employment shares in war-related sectors as the treatment. As shown by Rhode (2003), the concentration of sectors such as automobile factories, ship and boat building, wagon and carriage factories, and air transportation strongly predicts the assignment of war supply contracts. Again, as shown in Table A.4, we find similar results to those in the baseline specification.

Third, we test whether the long-run effects persist across different types of wartime contracts, including those less tied to specialized military production. We disaggregate total contract spending into three broad categories: (1) general-purpose and dual-use goods (food, textiles, chemicals, and miscellaneous); (2) basic intermediate inputs (metals); and (3) specialized war-related equipment (machinery, electronics, and transportation). As shown in Table A.5, the coefficients remain statistically significant and economically meaningful across all three categories, particularly in smaller initial manufacturing locations. The positive long-run effects for smaller locations are not confined to the sectors most exposed to postwar military contraction (machinery, electronics, and transportation) but appear in the food, textiles, and chemicals category as well. These results suggest that our findings are unlikely to be driven by postwar shifts away from war-specific sectors. Instead, they point toward broader, size-dependent agglomeration forces that operate across the manufacturing spectrum.

Postwar Military Spending A final concern is that wartime contracts proxy for continuing postwar defence spending rather than for the temporary shock itself. While military spending dropped precipitously at the end of the war, it did not drop all the way

to the prewar level, and it could be that locations with high wartime spending disproportionately benefited from this continued spending. To address the concern, we find that the correlation between contract spending and defense spending in later, postwar periods (1966 or 2000) at state level is low and insignificant, as shown in Table A.6.[11]

4 Quantitative Model and Structural Estimation

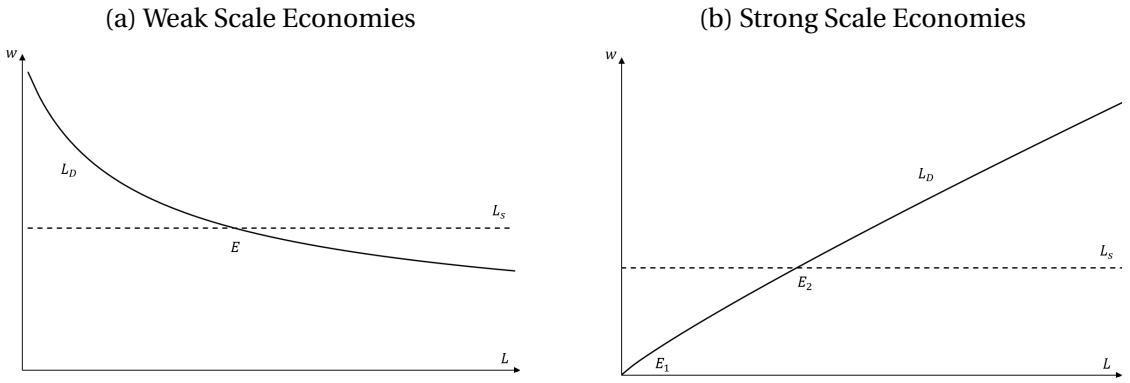
4.1 Quantitative Model

We build on the recent framework of Allen and Donaldson (2020), which enriches the basic quantitative economic geography model with a dynamic structure that can accommodate multiple steady states and path dependence. We add a) multiple sectors (agriculture, manufacturing, services), b) intermediate goods, c) a government sector, and d) local external economies of scale in the manufacturing sector that may affect productivity with non-constant elasticity. The first three modifications help match the specific experiment in our setting or add quantitative realism, and our modeling of these features is standard. The fourth is motivated by our empirical findings, as we explain below.

Our structural estimation will show that external economies of scale that are diminishing in sector size beyond a threshold value are critical for matching the empirically observed difference in treatment effects across initial conditions. Some intuition for why this is the case can be gleaned from considering the case of a single competitive sector in partial equilibrium, with labor l the only factor of production, facing a flat labor supply curve and constant-elasticity-of-substitution (CES) sector-level product demand with elasticity $\sigma > 1$. The elasticity of the sector-level inverse labor demand curve is $\frac{(\sigma-1)\gamma(l)-1}{\sigma}$, where $\gamma(l)$ is the elasticity of productivity with respect to sectoral employ-

[11] We analyze this correlation at the state level because reliable military spending data are only available at the state level in postwar years from the 1960s (Nakamura and Steinsson, 2014).

Figure 3. Labor Market Equilibria with CES Scale Function

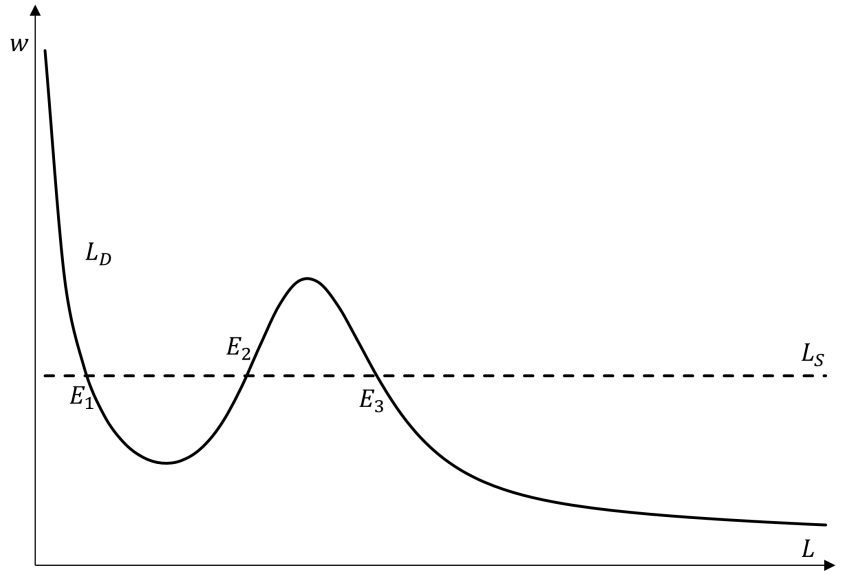


ment, which we assume is driven by external economies. When $\gamma(l)$ is constant, then (ignoring the case of equality) either $\gamma(l) < \frac{1}{\sigma-1}$ and the inverse labor demand curve is always downward sloping (panel (a) of Figure 3) or $\gamma(l) > \frac{1}{\sigma-1}$ and the inverse labor demand curve is always upward sloping (panel (b) of Figure 3).

In Panel (a) there is a unique equilibrium at point E that is interior and stable, and there is no poverty trap. In Panel (b) there are two equilibria, one stable at zero employment at E_1 and an unstable interior equilibrium at E_2. The equilibrium at E_1 could be described as a poverty trap, since a temporary shock large enough to carry employment past the point E_2 would result in a permanently higher level of employment and productivity. However, this model predicts that locations with *lower* initial employment should be *less* responsive to positive demand shocks in the long run than large places, the opposite of our empirical findings. More generally, this model is too stylized for quantitative work since the only two long-run outcomes are an employment level of zero or infinity, since it lacks partial equilibrium mechanisms to limit the growth of the sector.

In contrast, consider an S-shaped scale function that is initially flat in the sector size, increases rapidly over some range, then flattens again as the sector grows large. Given downward sloping product demand, this type of scale function can generate a sectoral

Figure 4. Labor Market Equilibria with S-shaped Scale Function



inverse labor demand curve that crosses the labor supply curve three times in the interior, implying a stable "low" equilibrium E_1 and a stable "high" equilibrium E_3, with an unstable equilibrium between them at E_2 (Figure 4). A temporary shock that moves the economy beyond the employment level at E_2 will eventually move the sector from the low to the high equilibrium, which then becomes self-sustaining; however, an economy that is initially in the high equilibrium always returns to the same steady state in response to a positive shock. This framework thus naturally implies the type of differential long-run responses to temporary shocks based on initial conditions that we found in our data analysis.

This analysis is simplistic in that it ignores general equilibrium considerations. The constant-elasticity model of Allen and Donaldson (2020) is capable of generating multiple stable interior steady states in general equilibrium. However, we find that the partial equilibrium prediction of extreme sectoral allocations still manifests to a large degree in general equilibrium, at least in our calibration. Thus, the constant elasticity specification tends to imply that either no location is in a poverty trap or almost every location is in a poverty trap. This is in conflict with the empirical evidence from Section 3. In addi-

tion, constant elasticity parameterizations that feature poverty traps also tend to imply that the ultimate end state of the economy is a very highly (and counterfactually) spatially concentrated equilibrium (a "black hole" in the terminology of Fujita, Krugman, and Venables (2001)). For these reasons, our estimation procedure strongly prefers scale economies that are sufficiently diminishing with large sector sizes. We discuss this issue further in Section 4.3.

While the quantitative literature in trade and economic geography has primarily focused on models in which the scale function is constant elasticity, this choice is typically made for reasons of analytical convenience and does not have a strong theoretical foundation *per se*.[12] Mechanisms that can naturally generate an approximately S-shaped relationship between scale and productivity include aggregate demand spillovers (Murphy et al., 1989), technology adoption (Ciccone and Matsuyama, 1996; Ciccone, 2002; Buera et al., 2021), imperfections in credit markets (Banerjee and Newman, 1993; Galor and Zeira, 1993) and other mechanisms surveyed in Azariadis and Stachurski (2005).[13] We lack the data to study the specific mechanisms at work during our sample period, and hence we remain agnostic as to the ultimate sources of the scale effects at work.[14]

Setup Consider $i \in \{1, ..., N\}$ locations and discrete time $t \in \{0, 1, ...\}$. Individuals live for two periods. In the first period ("childhood"), they are born in their parents' location and consume the same as their parents. At the beginning of the second period ("adulthood"), they realize their own preferences and choose their residence as adults.

[12] The most common microfoundations of scale economies come from idea-based models such as Krugman (1980), Eaton and Kortum (2001) and Melitz (2003), which generate non-constant scale elasticities except under specific functional form assumptions.

[13] See also Garg (2025) for a recent quantitative empirical study of coordination failures that also uses a flexible functional form for scale economies.

[14] Our discussion has focused on non-linearities in the labor demand curve arising from non-constant scale elasticities ("agglomeration forces"), but in principle systematic non-linearities in the labor supply curve ("congestion forces") could also be a source of multiple interior equilibria. This possibility is not very well motivated by the existing theoretical and empirical literature, and we lack the data to rigorously distinguish between these two potential sources of non-linearity. Our model of congestion forces, idiosyncratic location preferences, is standard in the literature.

In this location, they supply a unit of labor inelastically, consume, and have ζ_t children apiece.

Production There are two tradable sectors, "agriculture" and "manufacturing," and a non-tradable sector, "services." Each location produces a differentiated variety of agricultural and manufacturing goods. The production in agriculture and services uses labor with constant returns to scale,

$$Q_{A,i,t} = T_{A,i,t} l_{A,i,t}, \tag{4}$$

$$Q_{S,i,t} = T_{S,i,t} l_{S,i,t}, \tag{5}$$

where T_A and T_S are exogenous productivities, and l_A and l_S are labor inputs. Manufacturing uses labor and intermediate inputs from the manufacturing sector. Producers perceive constant returns, but external economies of scale exist in total manufacturing labor. Manufacturing output is given by

$$Q_{M,i,t} = T_{M,i,t} f\left(l_{M,i,t-1}\right) l_{M,i,t}^{\alpha} q_{M,i,t}^{1-\alpha}, \tag{6}$$

where T_M is exogenous productivity, $f(l)$ is the endogenous productivity that depends on lagged sector scale, l_M is labor input, q_M is a bundle of intermediate inputs, and α is the labor share.

Government Government purchases manufacturing output from each location and throws it away. Aggregate government consumption is

$$G_t = \sum_i g_{i,t}, \tag{7}$$

where $g_{i,t} = \kappa_{i,t} G_t$ are government purchases from location i, and $\kappa_{i,t}$ is the purchase share. To fund these purchases, the government imposes a per-person lump-sum tax, ι_t, satisfying $\iota_t L_t = G_t$, where L_t is total population at time t.

Demand Preferences are given by a nested CES utility function, with an upper tier across sectors and a lower tier across locations within sectors,

$$C_{i,t} = \left(\sum_{s \in \{A,S,M\}} \rho_s^{\frac{1}{\eta}} C_{s,i,t}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \tag{8}$$

$$C_{s,i,t} = \left(\sum_{n \in N} C_{s,i,n,t}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \tag{9}$$

where $C_{s,i,n,t}$ is the consumption in location i of the sector s good produced in location n at time t, ρ_s is the sector-specific preference parameter, and η and σ are upper- and lower-tier elasticities of substitution. The lower tier elasticity of substitution σ is assumed to be the same for each sector, as well as for intermediate goods demand in the manufacturing sector. Let

$$\omega_{s,i,t} = \rho_s \left(\frac{P_{s,i,t}}{P_{i,t}} \right)^{1-\eta}, \forall s \in \{A, S, M\}, \tag{10}$$

where $\omega_{s,i,t}$ is the share of total expenditure $E_{i,t} = w_{i,t} L_{i,t} - \iota_t L_{i,t}$ on goods from sector s, $P_{s,i,t}$ is the sector s price index, and $P_{i,t}$ is the overall price index. Then expenditure on each sector is given by

$$E_{s,i,t} = \omega_{s,i,t} \left(w_{i,t} L_{i,t} - \iota_t L_{i,t} \right), \forall s \in \{A, S, M\}, \tag{11}$$

where $L_{i,t}$ is the total population in location i and time t.

Goods Market Clearing and Trade There are three sources of demand for each good: final demand by consumers, intermediate demand, and final demand by the government. Manufactured goods trade incurs iceberg costs $\tau_{i,n,t}$. Under perfect competition in input markets, total manufacturing wages are proportional to total sales,

$$w_{i,t}l_{M,i,t} = \alpha \left[c_{M,i,t}^{1-\sigma} \left(\sum_{n \in N} \tau_{i,n,t}^{1-\sigma} \frac{E_{M,n,t} + \frac{1-\alpha}{\alpha} w_{n,t} l_{M,n,t}}{P_{M,n,t}^{1-\sigma}} \right) + \kappa_{i,t} G_t \right], \tag{12}$$

where $c_{M,i,t} = \frac{w_{i,t}^{\alpha} P_{M,i,t}^{1-\alpha}}{f(l_{M,i,t-1}) T_{M,i,t}}$ is proportional to the unit cost in manufacturing and $P_{M,j,t} = (\sum_i (\tau_{i,j,t} c_{M,i,t})^{1-\sigma})^{\frac{1}{1-\sigma}}$ is the intermediate goods price index. Agricultural goods are freely traded, and their market clearing conditions are

$$w_{i,t}l_{A,i,t} = \left(\frac{w_{i,t}}{T_{A,i,t}} \right)^{1-\sigma} P_{A,t}^{\sigma-1} \sum_{n \in N} E_{A,n,t}, \forall i, t, \tag{13}$$

where $P_{A,t} = (\sum_i (c_{A,i,t})^{1-\sigma})^{\frac{1}{1-\sigma}}$ is the price index for agricultural goods, with unit costs $c_{A,i,t} = \frac{w_{i,t}}{T_{A,i,t}}$. Finally, market clearing in the services sector yields

$$w_{i,t}l_{S,i,t} = E_{S,i,t}, \tag{14}$$

with $P_{S,i} = c_{S,i} = \frac{w_{i,t}}{T_{S,i,t}}$.

Migration Adults in i at $t-1$ have ζ_t children. At time t, children make their migration choice to maximize their adult welfare. Utility depends on location-specific amenity $H_{i,t}$, migration frictions $\xi_{i,n,t}$, and idiosyncratic preference $\epsilon_n(\ell)$,

$$U_{i,n,t}(\ell) = H_{n,t} \xi_{i,n,t} \frac{w_{n,t} - \iota_t}{P_{n,t}} \epsilon_n(\ell) \equiv U_{i,n,t} \epsilon_n(\ell). \tag{15}$$

We assume that $\epsilon_n(\ell)$ is distributed Fréchet with shape parameter θ, and thus the probability of choosing n is $\pi_{i,n,t} = P(U_{i,n,t} \geq \max_{j \neq n} U_{i,j,t})$. Note that $P(U_{i,n,t}(\ell) \leq u) =$

$\exp(-U_{i,n,t}^{\theta} u^{-\theta})$. The migration flow from i to n at time t is

$$L_{i,n,t} = \zeta_t \frac{U_{i,n,t}^{\theta}}{\sum_{j \in N} U_{i,j,t}^{\theta}} L_{i,t-1}. \tag{16}$$

Dynamic Equilibrium

Definition 1. *Given initial conditions* $\{L_{n,0}, l_{M,i,0}, l_{A,i,0}\}$*, an equilibrium is a sequence of wages and labor allocations such that*

1. *Total sales are equal to payments to labor in each sector as given by equations* (12) *and* (13)*.*

2. *Government budget balance is satisfied.*

3. *Migration flows are given by equation* (16)*.*

4. *Local and global labor markets clear:*

$$\begin{aligned} L_{i,t} &= l_{M,i,t} + l_{A,i,t} + l_{S,i,t}, \forall i, t, \\ \sum_{n \in N} L_{n,t} &= L_t = \zeta_t L_{t-1}, \forall t. \end{aligned}$$

The dynamics of the model are driven by costly migration and by the lagged effect of scale on productivity. Because the scale effects operate with a lag and there is no forward looking behavior, the within-period equilibrium conditions at time t are the same as those of a static model without scale effects, and hence we expect that there is always a unique equilibrium path from every initial allocation.[15] However, depending on the parameter values there may be multiple long-run steady states depending on the initial allocations (Allen and Donaldson, 2020).

[15] Rigorous theoretical conditions for unique equilibria have not been established for this type of static multisector economic geography model without scale economies. However, both theoretical priors and practical experience suggest that it is reasonable to expect uniqueness.

4.2 Estimation Using Indirect Inference

We calibrate the majority of the parameters of the model to standard values in the literature (Table 4) and use indirect inference to estimate parameters of the external economies of scale function $f(l)$. As described earlier in the section, we conjecture that matching the empirically observed difference in treatment effects across locations according to the initial size of the manufacturing sector may require $f(l)$ to diminish in sector size. We parameterize the scale function as

$$f(l) = \begin{cases} 250^{\gamma} & l_{Mit-1} \leq 250, \\ l_{Mit-1}^{\gamma} & 250 < l_{Mit-1} < l_{*}, \\ l_{*}^{\gamma} & l_{Mit-1} \geq l_{*}, \end{cases} \tag{17}$$

where γ is the slope parameter and l_* is the location parameter. This function approximates an S-curve in manufacturing employment: initially flat, increasing over some intermediate range, and flat again after local manufacturing scale exceeds l_*.[16] This is a simple two-parameter family that nests the standard constant elasticity case above 250 workers as the location parameter goes to infinity, yet is flexible enough to match the empirical moments.

We estimate the slope and location parameters using classical minimum distance, sometimes referred to as "calibration with standard errors." We minimize the distance between the empirical ATTs for long-term manufacturing employment, for initially low and high manufacturing locations (column 2 of Table 2) and the corresponding mo-

[16] The initially flat segment plays a dual role. First, it ensures that the marginal revenue product of labor in manufacturing tends to infinity as local manufacturing employment tends to zero in each county, just as in a model with no scale economies, ensuring positive manufacturing employment in every location. Second, it avoids placing weight on the large number of counties with very small reported manufacturing employment when calibrating the model. We suspect these entries are severely contaminated with measurement error. Our results are robust to varying the cutoff for the initial flat segment.

ments generated by the model,

$$Loss = \left(\beta^{model} - \beta^{data}\right)\left(\beta^{model} - \beta^{data}\right)'. \tag{18}$$

We generate the model moments using the following procedure. First, we guess the parameters of the scale function. Next, we invert the model to obtain the exogenous local sectoral total factor productivities (TFP) and amenities that rationalize the observed equilibrium in 1940. Then we shock the model with the empirically observed temporary local war spending shocks κ_i for each location simultaneously, and solve forward for the steady state holding all exogenous TFPs and amenities constant at their 1940 levels, and with population growth set to zero. This establishes the "treatment" scenario for each location. Then, for each treated location i, we return to the observed equilibrium in 1940 and set $\kappa_i = 0$, but retain all treatments in other locations at their observed values and solve the model forward again to obtain the control observation for i, i.e. the outcome for county i in the absence of treatment in county i, all else equal. Finally, we compute the treatment effect for i as the difference between county i's manufacturing employment in the two scenarios; averaging across treated counties in each group (initially high or low manufacturing employment) gives the model-implied ATT for each group (β^{model}).

We systematically search the parameter space by repeating this procedure for different pairs of parameters (scale and location) using a combination of global and local methods. Once we arrive at the pair that minimizes equation (18), we compute standard errors for the slope and location parameters (collected in the vector ε) using the delta method (Newey and McFadden, 1994) ,

$$Z = (G'G)^{-1}G'VG(G'G)^{-1}, \tag{19}$$

where Z is the variance-covariance matrix of $\hat{\varepsilon}$, V is the variance-covariance matrix of

the empirical moments (computed via block bootstrap), and $G = \partial\beta^{model}/\partial\varepsilon'$, which we compute numerically.

Since our estimation procedure is based on comparing steady states with identical exogenous parameters that differ only by their history, by construction any parameter combination that does not exhibit long-run path dependence in response to the empirical shocks will have $\beta^{model} = 0$, which is far from matching the empirical moments. Thus estimation is a matter of finding a parameter vector that generates the correct empirically observed pattern of path dependence, one that primarily affects counties with initially low manufacturing employment. Speaking loosely, the size of the scale elasticity γ is almost exclusively responsible for the existence or non-existence of path dependence in the model, while the location parameter l_* primarily controls which counties exhibit path dependence in response to shocks. Both parameters influence the size of the long-run ATT for those counties that exhibit path dependence, which also depends on the empirical distribution of the shocks.[17]

To invert the model we require data on $\{w_{i,t}, l_{A,i,t}, l_{M,i,t}, l_{S,i,t}, G_t, \kappa_{i,t}\}_{t\in\mathcal{T}}$, where $\mathcal{T} = \{0, 1, ..., T\}$, and knowledge of some parameters.[18] We calibrate the trade and migration elasticities to $\sigma = 8$ and $\theta = 3.3$, and the labor share to $\alpha = 0.5$. We set $\eta = 0.33$ and set the sectoral shares ρ_s such that aggregate labor supply adds up to aggregate labor demand in each sector. While these do not vary over time for our structural estimation described above in Section 4.2, they will for some counterfactual scenarios described below in Section 5, in which we calibrate their changes over time from the postwar data. Similarly, while our structural estimation sets population growth to zero, for other coun-

[17] Online Appendices D and E report robustness checks where we relax both the number of parameters and the number of moments. Intuitively, our two-parameter family is adequate because our empirical moments are informative regarding the size range in which locations experience long-run effects and the average size of the long-run effect, both of which are controlled by our two parameters. Additional parameters, for example allowing the right tail of the scale function to have non-zero slope, are not very well identified because locations that are on that right tail do not experience long-run effects in either the model or the data. Identifying a more flexible functional form would thus require also using the short-run responses for identification, and is beyond the scope of this paper.

[18] See Online Appendix C for details of the model inversion.

TABLE 4. Summary of calibration

Parameter	Notation	Value	Source
Migration elasticity	θ	3.3	Monte, Redding, and Rossi-Hansberg (2018)
Trade elasticity	σ	8	Eaton and Kortum (2002)
Manufacturing input share	α	0.5	Valentinyi and Herrendorf (2008)
Elasticity of substitution across sectors	η	0.33	Comin, Lashkari, and Mestieri (2021)
Sectoral share parameter	ρ_s	See notes	Add up to 1
Population growth	ζ_t	See notes	Population data
Slope parameter in scale function	γ	0.19	Indirect inference
Location parameter in scale function	l_*	1445	Indirect inference
Calibrated parameter in scale function	.	250	See notes

Notes: Sectoral share parameters are $\rho_{A,t} = [0.142, 0.094, 0.064, 0.035, 0.029, 0.023, 0.029, 0.024]$ and $\rho_{S,t} = [0.049, 0.040, 0.077, 0.129, 0.184, 0.250, 0.379, 0.572]$. Population growth is $\zeta_t = [1.208, 1.067, 1.156, 1.128, 1.209, 1.234, 1.224, 1.290]$. The first location parameter of the scale function is calibrated based on the average manufacturing share across counties multiplied by the 10th percentile of population in 1940. Note that $\rho_{A,t}$, $\rho_{S,t}$ and ζ_t only vary by time for the counterfactual analysis in Section 5, but not for the structural estimation.

terfactuals we set population growth as $\zeta_t = \frac{\sum_i L_{it}}{\sum_i L_{it-1}}$ using the observed data over the postwar time period. Finally, we set trade costs as

$$\tau_{i,n,t} = \begin{cases} distance_{in}^{\frac{1}{\sigma-1}} & i \neq n, \\ 1 & i = n, \end{cases} \tag{20}$$

and migration frictions as

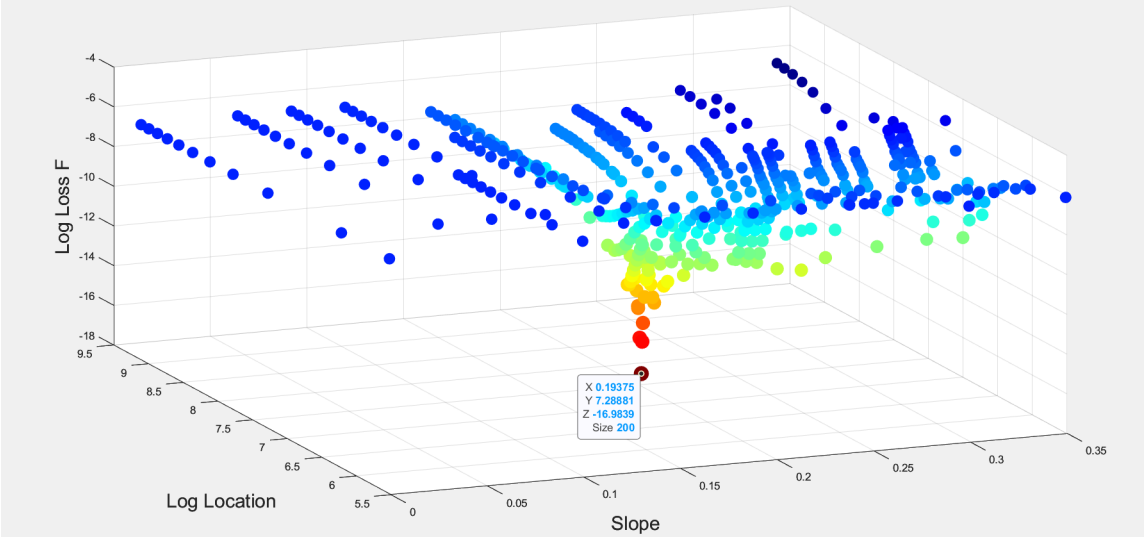
$$\xi_{i,n,t} = \begin{cases} distance_{in}^{-\frac{1}{\theta}} & i \neq n, \\ 1 & i = n, \end{cases} \tag{21}$$

where the distance between county i and n is defined in miles.[19] Table 4 displays our calibrated parameters. Table A.7 displays the variables we require for the model inversion along with their data counterparts and the sources.

The data used for simulation come from the same sources as those described in Section 2.1. Since we need wage information to invert the structural model, we impute

[19]Trade costs are identified only up to scale: scaling the trade-cost matrix by a constant leaves trade shares unchanged and is absorbed in the inversion. We normalize the diagonal to one, as for migration costs.

Figure 5. Unrestricted Loss Function



Notes: This graph plots the loss (equation (18)) as a function of the slope and location parameters.

missing wages for the simulation sample using a dynamic panel regression specification. The analysis is based on a harmonized county-level data set of war supply contract values, manufacturing employment, and wages for the decadal years from 1900 to 2000. Additional details are documented in Online Appendix B.

4.3 Results

Figure 5 plots the loss as a function of the slope and location parameters. We achieve a loss that is numerically indistinguishable from zero at the minimum, matching the empirical estimates exactly up to the third decimal place. Both local and global identification is fairly strong in the depicted region of the parameter space, which we verify also holds outside the depicted region.[20]

Table 5 shows our parameter estimates and standard errors. We estimate a slope parameter $\gamma = 0.19$ and a location parameter $l_* = 1445$.[21] The location parameter is around

[20] See Online Appendices D and E for additional exercises assessing model fit.

[21] The estimated $\gamma = 0.19$ exceeds the cutoff of $\frac{1}{\sigma-1} = 0.143$, so it sits in the strong-scale-economy range that can generate multiple interior equilibria.

TABLE 5. Indirect Inference

Parameter	Estimate
Slope: γ	0.19
	(0.02)
Location: l_*	1445
	(1183)

Notes: See text for details.

the 66th percentile of the county manufacturing employment distribution in 1940. Our estimates therefore imply that manufacturing productivity grows rapidly, about 40% in total, as a county moves from the 10th percentile to the 66th percentile of the manufacturing employment distribution, with no growth in productivity for further increases in the manufacturing labor force.

It is difficult to make a clean comparison of our estimate of γ to those in the literature. First, we are not aware of any estimates of a local scale function at this level of geographic and sectoral disaggregation. Second, estimation of similar functions, for example scale elasticities in manufacturing at the country level (Lashkaripour and Lugovskyy, 2023; Bartelme et al., 2025), assume a constant elasticity function which is not comparable to ours. Third, as is common in the quantitative literature, our estimation is sensitive to other details of the model calibration such as the value of the trade elasticity and our suppression of other potentially relevant agglomeration or congestion forces. Rather than focus on the specific estimated scale function, which is after all only an approximation that may be highly context-specific, we find it more illuminating to focus on how the estimated scale function fits with the rest of the model to match the observed empirical treatment effects.[22]

Due to its prominence in the literature, it is interesting to evaluate the ability of the constant elasticity model to match the empirical moments. Figure 6 plots the log loss function (left axis) against the scale elasticity, with the location parameter restricted to

[22]With that said, our value of γ fits comfortably in the range of manufacturing scale elasticities at the country level estimated in Lashkaripour and Lugovskyy (2023) and Bartelme et al. (2025).

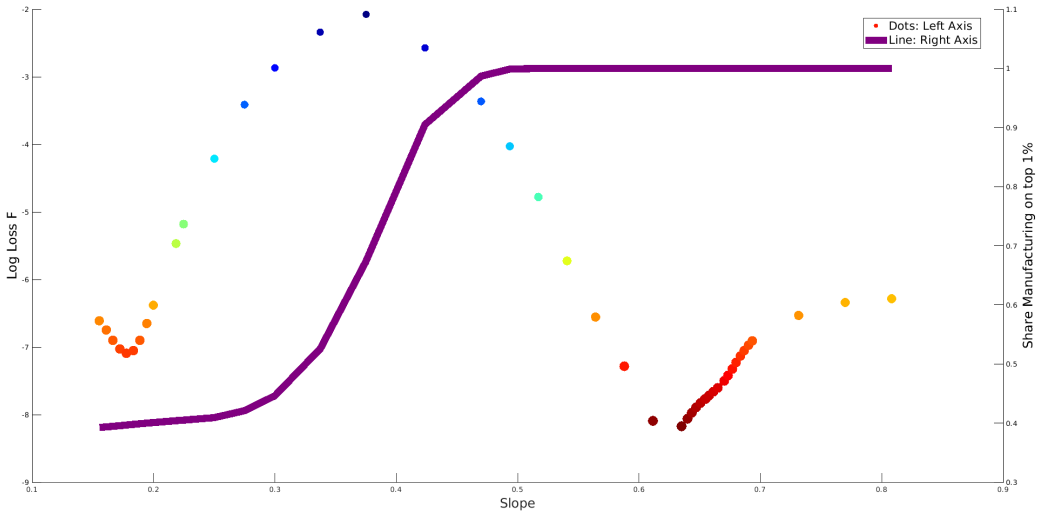
be infinite (the constant elasticity case). This restricted log loss function has a local minimum at around $\gamma \approx 0.19$, which is our unrestricted estimate, and a global minimum at $\gamma \approx 0.6$. The fit, however, is very poor compared to what we achieve with a finite location parameter; indeed, the lowest loss in Figure 6 is about the same as the highest loss displayed in Figure 5. The nature of the long-run equilibrium as γ increases beyond 0.4 or so in the constant elasticity model is "black hole"-like. Figure 6 also plots the share of manufacturing accounted for by the top 1% of counties in the steady state, which moves from being close to the actual 1940 value of about 40% to about 100% when $\gamma > 0.4$. In fact, for larger values of γ our simulations show that 3 counties (Monroe County, MS; Gibson County, TN; and Dinwiddie County, VA) account for essentially all of the nation's manufacturing employment in the steady state.[23] The constant elasticity model is thus unable to come close to matching the estimated heterogeneity in the long-run impact of WWII spending and, at least in this setting, the domain over which scale economies that are strong enough to generate path dependence is limited.

5 Counterfactual Analysis

We use our estimated model to re-examine the impact of the temporary component of WWII spending on the long-run evolution of U.S. economic geography. There is a sizable literature in history and economics on the topic, with some arguing that the WWII spending catalyzed long-run manufacturing growth in the South (e.g., Wright, 1986; Hooks and Bloomquist, 1992; Bateman et al., 2009) or the Pacific Coast (e.g., Rhode, 2003). Others have pointed out that the manner in which spending was allocated, primarily towards counties with a large established manufacturing base, could have had the opposite effect of reinforcing existing patterns of specialization (Fishback and Cullen, 2013; Rhode et al., 2018; Fishback and Jaworski, 2016). We do not directly address this

[23] In contrast, at our optimal parameter estimates the top 1% share in the steady state is around 40%, as in the data.

Figure 6. Restricted Loss Function



Notes: The dots plot the loss in equation (18) as a function of the slope parameter, assuming an infinite location parameter. The solid line plots the share of manufacturing in the top 1% of counties at the steady state.

historical debate, which considers the effects of both temporary and permanent (or very long-lived) changes in government spending associated with WWII, but focus solely on understanding the long-run effects that are attributable to the temporary component of WWII. Specifically, we compare the actual treated world given by the data from the years 1930-2000 to a counterfactual world in which the WWII spending did not occur, but all other exogenous variables evolved as they did in the data. In order to capture long-run outcomes, we focus our comparison on the actual and counterfactual equilibria in the year 2000.

Table 6 highlights the key finding: while the temporary WWII spending had a sizable long-term impact on some individual counties, the effects were not consequential at the aggregate level. Columns 1–2 show the weighted and unweighted average manufacturing shares across counties in the two scenarios, which are similar. The average absolute difference in manufacturing employment (column 3) is a bit over 1%, indicating that at least some counties experienced long-term effects on manufacturing employment.

TABLE 6. Aggregate Differences in Manufacturing Growth in 2000

	(1) Agg. Manuf. Share	(2) Avg. Manuf. Share (County)	(3) Avg. Absolute Value of % Difference	(4) Rural Share Manuf. Emp.	(5) % Change in Avg. Real Income	(6) % Change in Avg. Ex-ante Welfare
Data	0.1227	0.1380	.	0.1107	.	.
Counterfactual	0.1227	0.1376	0.0124	0.1103	0.00045	0.00020

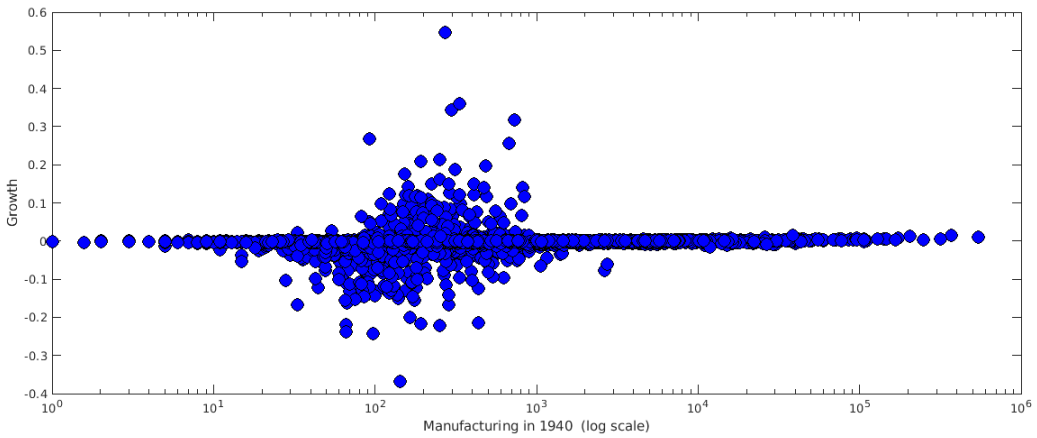
Notes: This table depicts aggregate statistics. Real income is wages divided by the price index. Ex-ante welfare is given by $\Phi_{i,t} = \left(\sum_{j \in N} U_{i,j,t}^{\theta}\right)^{1/\theta}$. The last two columns show the average change in real income and ex-ante welfare per person.

Column 4 shows no major shifts in the concentration of manufacturing in rural areas. Finally, columns 5–6 depict no substantial effect on real income per person and ex-ante welfare per person.

We further examine the impact at the individual county level. Figure 7 plots the log difference in manufacturing employment between actual and counterfactual for each county, against the initial manufacturing employment in 1940. Unsurprisingly, the differences are concentrated in counties with under 1500 manufacturing employees. Many differences are negative, implying that the long-run growth of manufacturing in some smaller counties came at the expense of other small counties where manufacturing shrank. The lack of an aggregate impact implies that positive and negative effects balanced out. While efficiency considerations would tend to dictate an overall reallocation of manufacturing activity to smaller counties and a reduction in the overall manufacturing employment share (due to the elasticity of substitution across sectors η being smaller than 1), the WWII spending shock did not catalyze much of a shift in this direction.

The map in Figure 8 highlights regional shifts in manufacturing employment. Overall, the Northeastern and Midwestern manufacturing belt gained manufacturing employment, consistent with the view that the WWII spending reinforced their regional comparative advantage. Most counties in the South experienced declines in manufac-

Figure 7. Long-term Impact of WWII Manufacturing Contracts on Manufacturing Employment: County-Level Differences



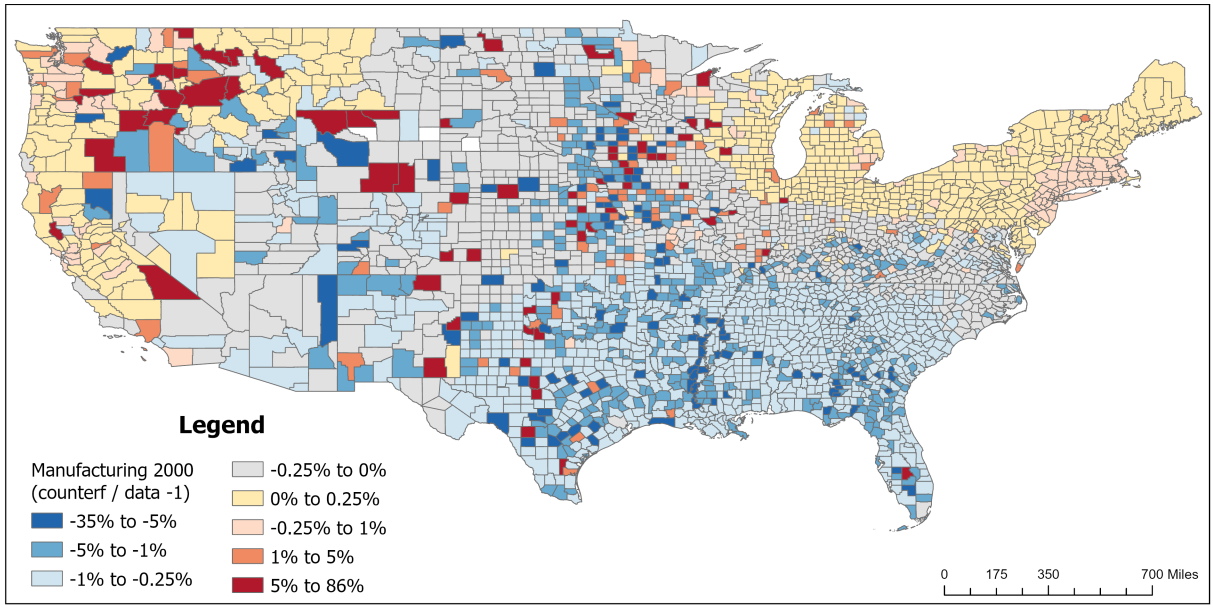
Notes: This scatter compares the outcome in the treated world versus the untreated world in 2000 across counties.

turing employment relative to the counterfactual world.[24] Regionally, the largest growth in manufacturing employment came in the Pacific Northwest, consistent with Rhode (2003), but even there the regional increase in manufacturing employment was small (0.5%). In all regions we see a rich pattern of impacts across counties, with some winners and some losers, even as regional outcomes are correlated on the whole. This reflects the balance between agglomeration economies at the county level, stemming from production externalities embodied in the scale function, and regional agglomeration economies due to the presence of trade costs and input linkages.

In sum, the counterfactual analysis reveals that while temporary WWII contracts had localized long-term impacts, prompting some underindustrialized counties to es-

[24]There is a rich literature studying the role of the changes in state and federal spending associated with the New Deal and WWII on economic development in the South, particularly in manufacturing. Historical studies that focus on long-lived spending changes and infrastructure improvements, and consider a range of quantitative and qualitative evidence, have generally argued that they helped catalyze a lasting change in the South's regional growth trajectory (e.g., Wright, 1986; Hooks and Bloomquist, 1992; Bateman et al., 2009). Econometric studies that focus on isolating specific shocks and comparing effects across counties have tended to find modest or zero effects (Jaworski, 2017; Jaworski and Yang, 2025), and our findings are similar. These two perspectives are not necessarily incompatible, as they emphasize different types of shocks and different levels of agglomeration forces (regional vs. local). A synthesis of these two perspectives would be an interesting area for future research.

Figure 8. Long-term Impact of WWII Manufacturing Contracts on Manufacturing Employment: Regional Map



Notes: This map compares the outcome in the treated world versus the untreated world in 2000 across counties.

cape local poverty traps, these effects did not translate into aggregate national gains. This lack of aggregate impact follows from the restricted domain of coordination failures, which bounds agglomeration forces. Because the S-shaped scale function flattens above 1,500 workers, once a localized big push drives a county past the threshold, additional labor inflows no longer yield further external economies of scale. Consequently, the long-run growth of manufacturing in successful treated counties primarily comes at the expense of other small counties below the treatment threshold. Since the limited domain of coordination failures bounds the capacity of spatial reallocation to generate net national productivity gains, the positive and negative local effects cancel out, leaving aggregate welfare little changed.[25]

[25] In principle, the lack of aggregate impact could also reflect the untargeted nature of the intervention, raising the possibility that some *other* temporary policy might have been more consequential for reallocation and welfare. While we cannot rule this possibility out completely due to the very large number of feasible policies, we did a number of counterfactual experiments with different policies that redistributed the WWII contracts more heavily towards smaller locations that might experience a big push. We were unable to generate large long-term aggregate reallocation or welfare effects from even substantial redistributions of contracts, suggesting that more targeted temporary policies would also have had

6 Conclusion

This paper studies the role of path dependence in regional development by investigating how initial conditions govern the persistence of large temporary shocks. By analyzing the long-term spatial effects of WWII supply contracts in U.S., we make two primary contributions. Empirically, we show that large-scale, temporary public investments generate permanent increases in population and manufacturing employment, but that these effects are highly state-dependent. Sustained industrial growth occurs exclusively in counties with low initial levels of manufacturing development, whereas heavily industrialized counties experience only transient gains that completely reverse in subsequent decades. Methodologically, we rationalize these findings by developing a quantitative multi-region economic geography model featuring path dependence and a non-constant elasticity of local scale economies.

Our quantitative re-evaluation of this historical episode offers a nuanced perspective on the significance of path dependence in economic development. On the one hand, we provide clear evidence that temporary policy interventions can overcome local coordination failures and permanently alter a region's long-run trajectory, validating the core mechanism of big push theories. On the other hand, our counterfactual simulations show that the aggregate national effects of this massive public spending shock on long-run manufacturing employment, welfare, and broader regional reallocation were remarkably limited. This divergence occurs because the restricted domain over which coordination failures apply bounds the capacity of localized reallocation to generate national productivity gains. Going forward, these results highlight the importance of having a framework that allows for flexibility in the scale of activity over which coordination failures operate.

limited long-run effects.

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Online Appendix

Initial Conditions and the Big Push

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October 2026

A Additional Tables and Figures

Figure A.1. Cumulative Probability Distribution of Initial Manufacturing Employment



Notes: This figure shows the cumulative probability distribution of manufacturing employment in 1930 in the sample.

TABLE A.1. Summary Statistics of Pre-WWII Regional Characteristics in 1930

	Treated Counties		Control Counties	
	Mean	Std. Dev.	Mean	Std. Dev.
Log population	10.428	0.596	10.031	0.520
Log manufacturing employment	7.105	1.286	6.138	1.316
Log employment	9.445	0.622	9.015	0.539
Agricultural employment share	0.359	0.180	0.486	0.192
Manufacturing employment share	0.138	0.106	0.094	0.095
Log average manufacturing wage	2.738	0.296	2.631	0.362
% Urban	0.359	0.203	0.216	0.195
Log average farm value	6.815	0.652	6.602	0.619
Log median housing values	10.683	0.382	10.462	0.391
Log median contract rents	10.045	0.414	9.834	0.407
% White	0.880	0.166	0.852	0.203
% Illiterate	0.047	0.048	0.057	0.059
% households owning radio	0.310	0.166	0.247	0.173
% Urbanization Rate	0.033	0.018	0.025	0.018
% Whites foreign born	0.051	0.056	0.043	0.053
Max elevation (meters)	1,675.159	2,084.700	1,761.862	2,292.476
Elevation range (max-min)	1,177.056	1,964.030	1,186.685	2,006.357
Area (squared miles)	780.856	846.258	769.884	691.974

Notes: Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms. To better balance treated and control samples, we drop counties whose pre-WWII characteristics yield extreme propensity scores of treatment higher than 99% or lower than 1%. All monetary values are in constant 2000 USD.

TABLE A.2. Contract Allocation and Initial Manufacturing Conditions

	(1) Treat	(2) ContractValue	(3) Treat	(4) ContractValue
Auto and Air Share	0.009	0.080		
	(0.004)	(0.024)		
Auto and Air Share Big			0.127	0.896
			(0.023)	(0.185)
Log Manuf. Emp. 1930	0.015	-0.017	0.025	0.074
	(0.058)	(0.565)	(0.060)	(0.596)
Controls	Yes	Yes	Yes	Yes
Obs.	1,777	1,777	1,780	1,780
Adj. R^2	0.377	0.530	0.382	0.532

Notes: This table shows the relationship between wartime contract allocation and initial manufacturing conditions in 1930 across counties. The dependent variables are treatment indicators and log contract value per 1930 manufacturing employee. Key independent variables are log manufacturing employment levels, predicted log contract value, and employment shares in sectors highly related to the war effort including automobile factories, ship and boat building, wagon and carriage factories, and air transportation. All regressions adjust for the control variables described in the text. Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms.

TABLE A.3. Robustness Check Controlling for Industry Composition

	(1) Pop 1930-2000	(2) ManufEmp 1930-2000	(3) Pop 1930-1960	(4) ManufEmp 1930-1960	(5) Pop 1960-2000	(6) ManufEmp 1960-2000
Treat*LowM	0.026	0.036	0.025	0.083	0.027	−0.004
	(0.008)	(0.010)	(0.008)	(0.020)	(0.009)	(0.010)
Treat*HighM	0.009	0.012	0.019	0.060	0.004	−0.022
	(0.008)	(0.009)	(0.009)	(0.015)	(0.008)	(0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows robustness checks controlling for employment shares in sectors highly related to the war effort including automobile factories, ship and boat building, wagon and carriage factories, and air transportation. All regressions adjust for the control variables described in the text. Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms.

TABLE A.4. Reduced-Form Estimation Based on Employment Shares in War-Related Sectors

	(1) Pop 1930-2000	(2) ManufEmp 1930-2000	(3) Pop 1930-1960	(4) ManufEmp 1930-1960	(5) Pop 1960-2000	(6) ManufEmp 1960-2000
War Share Big*LowM	0.059	0.067	0.068	0.104	0.052	0.025
	(0.015)	(0.015)	(0.017)	(0.034)	(0.015)	(0.017)
War Share Big*HighM	0.003	0.006	0.023	0.032	−0.009	−0.011
	(0.006)	(0.008)	(0.009)	(0.014)	(0.006)	(0.010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows reduced-form effects based on initial employment shares in sectors highly related to the war effort including automobile factories, ship and boat building, wagon and carriage factories, and air transportation. *War Share Big* is an indicator equal to one for counties whose employment share in war-related sectors is above the median, among counties with a positive share in those sectors. Standard errors are clustered at the state level.

TABLE A.5. Long-Term Impact on Growth Rate of Outcomes by Contract Types

	(1) Pop 1930-2000	(2) ManufEmp 1930-2000	(3) Pop 1930-2000	(4) ManufEmp 1930-2000	(5) Pop 1930-2000	(6) ManufEmp 1930-2000
Treat*LowM	0.020	0.037	0.053	0.075	0.042	0.058
	(0.006)	(0.008)	(0.019)	(0.019)	(0.014)	(0.013)
Treat*HighM	0.006	0.012	−0.000	0.002	0.001	0.006
	(0.005)	(0.007)	(0.007)	(0.008)	(0.006)	(0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Contract Type	General	General	Intermediate	Intermediate	Specialized	Specialized

Notes: Treated counties correspond to those that received WWII war supply contract value per 1930 manufacturing employee exceeds 8 in logarithmic terms. Treatment status is classified into three categories of contract spending as follows: (1) general-purpose and dual-use goods (food, textiles, chemicals, miscellaneous); (2) basic intermediate inputs (metals); and (3) specialized war-related equipment (machinery, electronics, transportation).

TABLE A.6. Relationship Between Post-WWII Military Spending and War Supply Contracts

	(1) MilitaryExp 1966	(2) MilitaryExp 2000	(3) MilitaryExp 1966	(4) MilitaryExp 2000	(5) MilitaryExp 1966	(6) MilitaryExp 2000
Contract value	0.002	0.027				
	(0.023)	(0.021)				
Treat			0.096	0.438		
			(0.413)	(0.380)		
Treat (median)					-0.093	0.271
					(0.404)	(0.393)
Obs.	50	50	50	50	50	50

Notes: This table shows coefficients of regressing post-WWII military spending in 1966 and 2000 at the state level on wartime contract spending or treatment status. Treatment status is defined based on whether the log contract value is above 8 (columns 3–4) or the median value (columns 5–6). Robust standard errors are reported in parentheses.

TABLE A.7. Data Sources for Structural Estimation

Variable	Source
Total employment, $L_{i,t}$	Decennial Census
Manufacturing employment, $l_{M,i,t}$	Decennial Census
Agricultural employment, $l_{A,i,t}$	Decennial Census
Wages, $w_{i,t}$	Decennial Census
Government expenditure in manufacturing, $\kappa_{i,t}$	Digitized WWII supply contracts from volumes originally issued by the Civilian Production Administration
Distance between locations (in miles)	NBER Distance Database

Notes: Our sample corresponds to the period 1900-2000.

B Data Used in Estimation and Simulation Samples

The data for WWII supply contracts come from volumes originally issued by the Civilian Production Administration (formerly the War Production Board), which record all defense contracts greater than $50,000 issued between June 1940 and September 1945. Our measure of government defense spending is the aggregate monetary value of war supply contracts in each county during that period.

The empirical analysis is based on county-level data for the years 1900 to 2000. Data on county characteristics are the same as those in Kline and Moretti (2014) and come from a variety of sources including the Census of Population and Housing and Agricultural Census. In particular, the data for the years 1900 to 1930 comes from the Historical, Demographic, Economic, and Social Data: The United States, 1790-2000, ICPSR 2896 (specifically Parts 20, 22, 24, 26, 29, and 85, corresponding to the 1900, 1910, 1920, and 1930 Censuses, as well as the 1930 Census Parts I and IV Families, and the 1910 Census of Agriculture). However, variables for manufacturing and agricultural employment were constructed from individual-level data sourced from a 1% sample extract of the 1900, 1910, 1920, and 1930 Censuses, available through IPUMS.

Variables for the period from 1940 to 1970 were developed using two primary data sources: the County and City Data Book [United States] Consolidated File: County Data, 1947-1977 (ICPSR 7736) and the Historical, Demographic, Economic, and Social Data: The United States, 1790-2000 (ICPSR 2896, specifically Parts 70, 72, 74, and 76, corresponding to the County Data Books for 1947, 1952, 1962, and 1972). The definitions of the variables were mostly consistent across these two data sources. In case of discrepancies, we use the definitions from ICPSR 2896. The data from 1980 to 2000 were derived from census extracts provided by the National Historical Geographic Information System.

The quality of some key variables in our dataset is imperfect, particularly at the early years of the sample period, where measurement error is likely substantial. County-level

data on workers' wages are not directly available in the initial periods. Following Kline and Moretti (2014), we proxy average manufacturing wages by dividing the total annual manufacturing wage bill by the estimated number of manufacturing workers in each county. However, even the wage bill data are incomplete or missing in some counties. For agriculture, no county-level wage bill data exist at all, precluding direct computation of average agricultural wages. Since the structural model requires wage observations for every location and period to invert labor demand and recover productivity shocks, we impute missing wages in the simulation sample using available information from nearby periods or county-specific trends. Specifically, we impute missing wage values for county i in decade t using a dynamic panel regression specification as follows:

$$w_{it} = \lambda w_{it-1} + \mathbf{b} X_{it} + \delta t + \mu_i,$$

where we control for lagged wage w_{it-1}, county characteristics X_{it} (which include population, median house value, median rent, ratio of white population, agricultural employment share, urban population share, when possible), county fixed effects μ_i, and a linear time trend.[26]

To avoid issues with changes in county definitions over time due to splits and merges, we use the crosswalk provided by Eckert et al. (2020) to obtain a sample of harmonized counties between 1900 and 2000. We also exclude counties in Alaska or Hawaii as well as underpopulated counties with population fewer than 1,000 in any decade in the sample period or manufacturing employment fewer than 10 in 1930 from the empirical analysis.

[26] If lagged wage is missing, we drop the first term from the regression equation.

C Model Inversion

We infer TFP in manufacturing, agriculture, and services ($T_{M,i,t}, T_{A,i,t}, T_{S,i,t}$), and amenities ($H_{i,t}$) using observed data on population, sectoral employment, and wages. Given the data $\{w_{i,t}, l_{M,i,t}, l_{A,i,t}, l_{S,i,t}, L_{i,t}, G_t, \kappa_{i,t}\}_{t\in\mathcal{T}}$, where $\mathcal{T} = \{0, 1, ..., T\}$, and parameters $\{\sigma, \eta, \rho_s, \theta, \alpha, \tau_{i,n,t}, \xi_{i,n,t}, \zeta_t\}_{t\in\mathcal{T}}$, we invert the model and solve for $\{T_{M,i,t}, T_{A,i,t}, T_{S,i,t}, H_{i,t}\}_{t\in\mathcal{T}}$ for each location, up to a normalization for each variable.

1. The first step is to solve for $c_{M,i,t}, c_{A,i,t}$, and $c_{S,i,t}$ up to a normalization. Given the data, $c_{M,i,t}, c_{A,i,t}, c_{S,it}$ solves the implicit equations:

$$\begin{aligned} w_{i,t}l_{M,i,t} &= \alpha\left[c_{M,i,t}^{1-\sigma}\left(\sum_{n\in N}\tau_{i,n,t}^{1-\sigma}\frac{E_{M,n,t}+\frac{1-\alpha}{\alpha}w_{n,t}l_{M,n,t}}{\sum_j c_{M,j,t}^{1-\sigma}\tau_{j,n,t}^{1-\sigma}}\right)+\kappa_{i,t}G_t\right], \forall i,t \\ w_{i,t}l_{A,i,t} &= c_{A,i,t}^{1-\sigma}P_{A,t}^{\sigma-1}\sum_{n\in N}E_{A,n,t}, \forall i,t \\ w_{i,t}l_{S,i,t} &= E_{S,i,t}, \forall i,t \end{aligned}$$

   where

$$\begin{aligned} E_{s,i,t} &= \omega_{s,i,t}\left(w_{i,t}L_{i,t}-\iota_t L_{i,t}\right) \\ \omega_{s,i,t} &= \rho_s\left(\frac{P_{s,i,t}}{P_{i,t}}\right)^{(1-\eta)}, \forall s\in\{A,S,M\} \\ \omega_{A,i,t} &+ \omega_{S,i,t}+\omega_{M,i,t}=1 \end{aligned}$$

   and ρ_s are chosen such that market clearing is assured in all sectors for any unit cost vector $\{c_{M,i,t}, c_{A,i,t}, c_{S,it}\}$. Note that this equation is homogeneous of degree zero in $\{c_{M,i,t}, c_{A,i,t}, c_{S,it}\}$, and that all other variables are known from the data. There is a unique solution (up to scale) of this equation, $\frac{c_{M,i,t}}{c_{M,1,t}}, \frac{c_{S,i,t}}{c_{S,1,t}}, \frac{c_{A,i,t}}{c_{A,1,t}}$.[27]

[27] Our main data sources are the Decennial Censuses. Since most of the contract expenditures happened between 1940 and 1950, and we do not observe any data point in those years, we assume $G_t = 0$ when backing out sectoral TFP.

2. The second step is to recover the price index $P_{M,n,t}^{1-\sigma} = \sum_j \frac{c_{M,j,t}^{1-\sigma}}{c_{M,1,t}^{1-\sigma}} \tau_{j,n,t}^{1-\sigma}$ and compute:

$$\begin{aligned}
\frac{T_{M,i,t} f(l_{M,i,t-1})}{T_{M,1,t} f(l_{M,1,t-1})} &= \frac{c_{M,1,t}}{c_{M,i,t}} \frac{w_{i,t}^{\alpha} \left(\sum_j \frac{c_{M,j,t}^{1-\sigma}}{c_{M,1,t}^{1-\sigma}} \tau_{j,i,t}^{1-\sigma} \right)^{\frac{1-\alpha}{1-\sigma}}}{w_{1,t}^{\alpha} \left(\sum_j \frac{c_{M,j,t}^{1-\sigma}}{c_{M,1,t}^{1-\sigma}} \tau_{j,1,t}^{1-\sigma} \right)^{\frac{1-\alpha}{1-\sigma}}}, \forall i,t \\
\frac{T_{S,i,t}}{T_{S,1,t}} &= \frac{w_{i,t}/c_{S,i,t}}{w_{1,t}/c_{S,1,t}}, \forall i,t \\
\frac{T_{A,i,t}}{T_{A,1,t}} &= \frac{w_{i,t}/c_{A,i,t}}{w_{1,t}/c_{A,1,t}}, \forall i,t
\end{aligned}$$

3. The final step is to recover exogenous amenities. Armed with previous solutions, we compute price index $P_{i,t}^{1-\eta} = \sum_s \rho_s \left(P_{s,i,t} \right)^{1-\eta}$ and use the migration equation:

$$\begin{aligned}
L_{n,t} &= \zeta_t \sum_i \frac{U_{i,n,t}^{\theta}}{\sum_{j \in N} U_{i,j,t}^{\theta}} L_{i,t-1}, \forall n,t \\
U_{i,n,t} &= H_n \xi_{i,n,t} \frac{w_{n,t} - \iota_t}{P_{n,t}},
\end{aligned}$$

which can be solved uniquely (up to scale) for $\frac{H_{n,t}}{H_{1,t}}$.

D Robustness of the Scale Function

In Section 4.2 we estimated a scale function with a slope parameter and a location parameter. We test the robustness of that specification by allowing for a second slope that activates once manufacturing employment rises above the location parameter. Estimating a three-parameter scale function jointly is impractical. Instead, for each parameter of the scale function, we fix the others at their estimated values and sweep the remaining one over a grid. We start by fixing the location and slope parameters at their estimated values and looking for the second slope that provides the best fit. We then set the second slope to that value and sweep over a grid for the location parameter. We update the location parameter, sweep over the first slope, and so on. At each point we report the two targeted moments and the loss in equation (18).

Specifically, this exercise requires a scale function in which the elasticity above the kink is free. We therefore generalize the specification in Section 4.2 to

$$f(l) = \begin{cases} 250^{\gamma} & l_{Mit-1} \leq 250, \\ l_{Mit-1}^{\gamma} & 250 < l_{Mit-1} < l_{*}, \\ l_{*}^{\gamma} \left(\dfrac{l_{Mit-1}}{l_{*}} \right)^{\gamma_2} & l_{Mit-1} \geq l_{*}, \end{cases} \tag{A.1}$$

where γ_2 governs the extent to which scale economies continue to operate above the kink. The specification we estimate in the main text is the special case $\gamma_2 = 0$, in which the scale function is flat above l_*.

Table A.8 collects the three sweeps. Figure A.2 displays them. Each row of the figure is one sweep. The left column plots the targeted moments. The right column plots the loss.

D.1 Scale Economies Above the Kink

Panel (a) of Table A.8 and Panels (a) and (b) of Figure A.2 report the sweep over γ_2, holding γ and l_* at their estimates. The loss is practically flat between 0 and 0.05. It takes the value 2.77×10^{-6} at $\gamma_2 = 0$, falls to 2.12×10^{-6} between 0.01 and 0.025, and returns to 2.76×10^{-6} at 0.05. The fit worsens at $\gamma_2 = 0.1$.

The loss is nonetheless lowest at $\gamma_2 = 0.025$. Since the improvement in fit is negligible, this already suggests that setting $\gamma_2 = 0$ is immaterial for our conclusions. To further test this, in the following sweeps, we set $\gamma_2 = 0.025$.

D.2 The Location Parameter

Panel (b) of Table A.8 and Panels (c) and (d) of Figure A.2 report the sweep over l_*, holding both slopes fixed. The loss is U-shaped and minimized at the estimate of the main text, about 1445.

D.3 The Slope Parameter

Panel (c) of Table A.8 and Panels (e) and (f) of Figure A.2 report the sweep over γ, holding the location parameter at its estimate. The loss is minimized at the same slope of the main text. Both targeted moments increase almost linearly in γ. They cross their empirical targets at the same point, $\gamma = 0.1875$.

Taken together, the three sweeps show that our estimates in the main text do not rest on the calibration of the slope for the upper segment. Relaxing it to $\gamma_2 = 0.025$ leaves the slope and location parameters at their values in the main text.

TABLE A.8. Identification Sweeps of the Scale Function

Panel (a): Upper-segment elasticity γ_2 ($\gamma = 0.1875$, $l_* = 1445.5$)

	β_{LowM}	β_{HighM}	Loss
$\boldsymbol{\gamma_2 = 0}$	$\mathbf{0.0388}$	$\mathbf{0.0085}$	$\mathbf{2.77 \times 10^{-6}}$
$\gamma_2 = 0.0025$	0.0389	0.0087	2.56×10^{-6}
$\gamma_2 = 0.0050$	0.0389	0.0088	2.38×10^{-6}
$\gamma_2 = 0.0100$	0.0390	0.0090	2.12×10^{-6}
$\gamma_2 = 0.0250$	0.0394	0.0095	2.12×10^{-6}
$\gamma_2 = 0.0500$	0.0396	0.0103	2.76×10^{-6}
$\gamma_2 = 0.1000$	0.0393	0.0119	5.13×10^{-6}

Panel (b): Location parameter l_* ($\gamma = 0.1875$, $\gamma_2 = 0.025$)

	β_{LowM}	β_{HighM}	Loss
$l_* = 1000$	0.0309	0.0049	7.74×10^{-5}
$l_* = 1200$	0.0357	0.0066	1.67×10^{-5}
$\boldsymbol{l_* = 1445}$	$\mathbf{0.0394}$	$\mathbf{0.0095}$	$\mathbf{2.11 \times 10^{-6}}$
$l_* = 1600$	0.0408	0.0108	8.48×10^{-6}
$l_* = 1800$	0.0417	0.0124	1.93×10^{-5}
$l_* = 2000$	0.0424	0.0138	3.35×10^{-5}

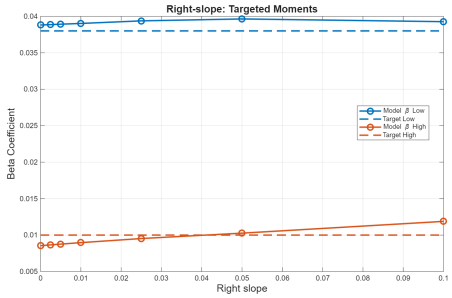
Panel (c): Slope parameter γ ($l_* = 1445.5$, $\gamma_2 = 0.025$)

	β_{LowM}	β_{HighM}	Loss
$\gamma = 0.1000$	0.0008	0.0002	1.48×10^{-3}
$\gamma = 0.1300$	0.0048	0.0010	1.18×10^{-3}
$\gamma = 0.1600$	0.0180	0.0040	4.35×10^{-4}
$\boldsymbol{\gamma = 0.1875}$	$\mathbf{0.0394}$	$\mathbf{0.0095}$	$\mathbf{2.12 \times 10^{-6}}$
$\gamma = 0.2100$	0.0620	0.0150	6.01×10^{-4}
$\gamma = 0.2400$	0.0857	0.0208	2.39×10^{-3}
Empirical target	0.038	0.010	

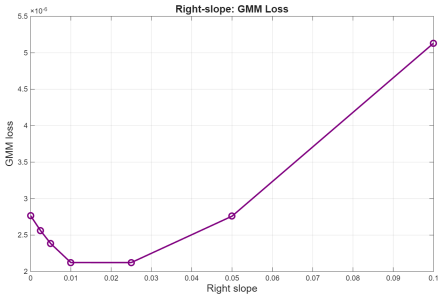
Notes: Each row re-solves the model in full at the indicated parameter value, holding the remaining parameters of the scale function fixed at the values given in the panel heading. β_{LowM} and β_{HighM} are the model-implied long-run ATTs on manufacturing employment for counties with initially low and high manufacturing employment, computed exactly as in the estimation. Loss is equation (18) against the empirical targets in the final row, taken from column (2) of Table 2. Bold rows denote the point estimate.

Figure A.2. Robustness of the Scale Function

(a) γ_2: targeted moments



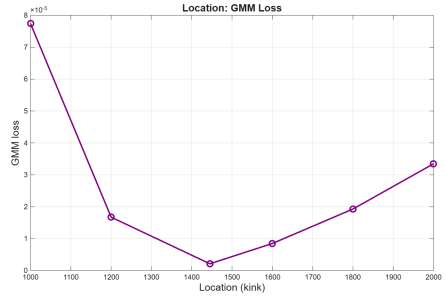
(b) γ_2: loss



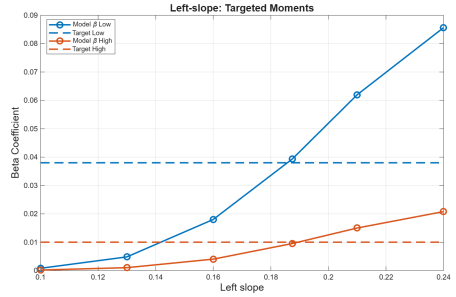
(c) l_*: targeted moments



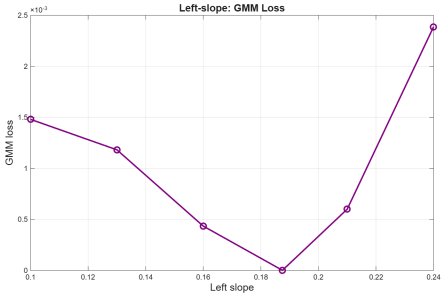
(d) l_*: loss



(e) γ: targeted moments



(f) γ: loss



Notes: Each row sweeps one parameter of the scale function over a grid, holding the other two fixed. The model is re-solved at every point. The top row sweeps the elasticity above the kink γ_2 over $[0, 0.1]$, holding the slope and location parameters at their estimates. The middle row sweeps the location parameter over $[1000, 2000]$, holding both slopes fixed. The bottom row sweeps the slope parameter over $[0.10, 0.24]$, holding the location parameter at its estimate. The middle and bottom rows hold $\gamma_2 = 0.025$. The left column plots the two targeted moments against their empirical targets, shown as dashed lines. The right column plots the loss in equation (18).

E Over-Identification: Four-Moment Matching

In this section we examine whether the estimated scale function reproduces treatment effects it was not asked to match. Because the estimation targets two moments with two parameters, the model is exactly identified. We use the four-way split of treatment effects reported in Table 3, which interacts initial manufacturing employment with the size of a county's war supply contracts. Large and small contracts are split at a log contract value of 9.36, as in the main text.

Table A.9 reports the results. The baseline column evaluates the model at our estimates. The re-fit column reports the model at the parameter pair that minimizes the equally weighted distance to all four moments. We compute it over the same solved grid used in the main text.

Once again, we see that within each contract size group, counties with initially low manufacturing employment display larger long-run responses than initially industrialized counties. This holds in both the data and the model. However, the model does not reproduce the contract size dimension. At our estimates it over-predicts the response of large-contract, low-manufacturing counties by a factor of more than two, 0.122 against 0.052 in the data. It under-predicts the response of large-contract, high-manufacturing counties by a factor of roughly three. Re-fitting the two parameters to all four moments improves the fit considerably, even though the location and slope parameters remain within the same order of magnitude.

The four-moment location parameter of 794 lies about 0.55 standard errors below our point estimate of 1445. Hence, the two are not statistically distinguishable.

TABLE A.9. Four-Moment Over-Identification Check

	Data	Baseline	Re-fit
Large contracts × LowM	0.052	0.1217	0.0502
Small contracts × LowM	0.029	0.0241	0.0265
Large contracts × HighM	0.022	0.0063	0.0012
Small contracts × HighM	0.001	0.0128	0.0064
Slope: γ		0.1875	0.2125
Location: l_*		1445	794
Loss		5.27×10^{-3}	4.70×10^{-4}

Notes: Data are the decadalized long-run effects on manufacturing employment from column (2) of Table 3. Large and small contracts are counties above and below a log contract value of 9.36, as in Table 3. Baseline evaluates the model at the estimates of Table 5. Re-fit reports the model at the parameters that minimize the equally weighted distance to all four moments over the solved parameter grid. Loss is the equally weighted squared distance between the four model and data moments.